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PROJECT REPORT

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Abstract

For the final year of undergraduate university degrees, students are often required to complete an individual final-year project to showcase the skills that they have learned throughout their years of study at university. When it comes to managing final-year projects and keeping on track with their project deliverables/tasks, both students and lecturers have a plethora of bespoke methods to track how well the project is coming along and for keeping up with the required project deliverables that need completing at various stages of the academic year. Some of these methods are more effective than others. Unfortunately, some students do not have an efficient way of self-managing their project and its deliverables and keeping up with the workload required to complete their project by its due date. Miscommunication between students and supervisors can occur in regard to the due dates of project deliverables and also what the requirements are for a project deliverable. Supervisors may also struggle to locate all of the different pieces of project work that students send to them if they do not have an efficient way of keeping track of their students' work. Creating a project deliverables management system (PDMS) with a web-based interface that is usable on both desktops and mobile devices would help both students and lecturers with the overall management of final-year projects. The system would provide a centralised location where all relevant information about the project and completed deliverables could be stored. The system will allow students to upload completed documents so that they are viewable to both student and their supervisor. The system will allow students and supervisors to communicate and discuss specific deliverables for the project right from within the PDMS, as well as provide reminders by email to both user types when a deliverable is due. The desired result of the PDMS is to increase the completion rate from students of project deliverables so that they can better manage their final-year projects and stay on track to have it completed.

1 - Introduction & Background

1.1 - Stakeholder Analysis

1.1.1 - Introduction

This project has specific stakeholders that will have some interest or involvement in the end solution. This section will examine specific stakeholders to clarify the project's impact on them. This section will also examine the distinct roles of the stakeholders in relation to the project's development and the required level of engagement needed to satisfy their needs. Additionally, this section will review the current problems that are faced by specific stakeholders and what the benefits are for solving these.

For this project, research was carried out to find who the specific stakeholders would be. The project client, who is one of the main stakeholders, was asked to give his opinions regarding this. After discussing possible stakeholders, four main stakeholders were identified. Once the stakeholders were identified, ways of engaging the specific stakeholder categories were considered based on their overall influence on the project.

1.1.2 - The Stakeholders

One of the stakeholders of this project is Dr Graeme Jones, lecturer, final-year project supervisor at Kingston University and the client of this project. Every year, Graeme helps to manage the final-year projects of several students in the computing school of the university. Graeme would like a website that can be used to manage the projects of his students and keep track of the individual project deliverables that are agreed between himself and individual students. Graeme would like to make sure that project deliverables are completed on time and to a good standard, so it is clear which students are on track to completing their projects.

Other stakeholders of this project would be final-year university students who are completing final-year projects/dissertations and also their project supervisors. The final stakeholders identified for this project are the parents of students. Parents would most likely want to ensure their children achieve high grades. Because of this, they will be interested in a project deliverables management system that helps their daughters/sons to organise their projects.

1.1.3 - Current Stakeholder Problems

Currently, students face challenges when trying to keep up to date with the regular completion of their project deliverables to a high standard. Supervisors also face the problem of managing all of the deliverables that due to be completed by students. The PDMS addresses these problems by providing both students and supervisors with a centralised location where projects and their specific deliverables can be managed.

1.1.4 - Benefits of Solving the Problems

Students

The students would benefit from this system because they would have a centralised location where they can manage their final-year project and all of the deliverables that need to be completed. The ability to see tasks for a project along with receiving reminders when a task is near its due date would help to reduce the recurrence of forgetting that something needs to be completed.

The Client & Supervisors

Supervisors would benefit from this system for reasons similar to that of the students. The PDMS would provide a centralised location in which supervisors could manage the individual projects of their students. It would also provide supervisors with the ability to manage specific deliverables for these projects. It would provide an easy way to communicate with students about their final-year projects too.

Parents

Parents would benefit from this system because they would know that their children will be competently managing their final-year projects by keeping up to date with the required tasks.

1.1.5 - How the Proposed Solution Relates to Existing Systems

An existing system that is somewhat similar to the PDMS is named Canvas. Canvas is a web and application-based tool used by many universities to track the submission of assignments form students. The software is also used for students to access relevant information from their modules, such as documentation provided by the relevant lecturer.

The proposed PDMS will differentiate itself from Canvas and provide additional value by having the software focus directly on final-year projects. The PDMS will allow students to do something that is not possible on Canvas which is to let students create project deliverables which the PDMS will be referring to as 'tasks' and then 'sub-tasks' within these tasks. When a user creates a task and simultaneously sets a due date, the student's supervisor will be able to see these tasks and both the student and supervisor will be sent reminders when the task is due to be completed. Supervisors will also have the ability to create tasks and subtasks for their students to complete too.

1.1.6 - Potential Challenges

Ironically, a challenge that could be faced when completing this project is self-managing the project deliverables for creating this system, along with the projects from other modules that development team will need to complete too.

Another potential problem for this project would be successfully integrating the required database to the webpage. There is a magnitude of errors that could occur, such as the webpage not being able to pull information from the database and correctly display this information to the webpage. This problem also works the other way in the sense that when entering data into the webpage, it may not successfully send back to the database.

1.2 - Aims & Objectives

1.2.1 - Aim

The overall aim of this project is to create a project deliverables management system (PDMS) with a web-based interface that is usable on both desktops and mobile devices. The PDMS aims to help both students and lecturers with the management of final-year projects. The system will need to provide a centralised location where all relevant information about a project and its relevant ongoing and completed tasks can be stored.

1.2.2 – Objectives

Database

One objective for this project is to create a database that will be used to store relevant user information such as account details of both students and supervisors. The database will also store the final-year projects of students and the deliverable tasks that students will complete for the project.

Goals

- For the development of the database, an ERD model will need to be created for the design of the database. The ERD can be made using applications such as StarUML which allows the creation of professional looking ERD database diagrams.
- A relational database model will need to be created for this project. The database model will be based on the ERD diagram that will be created prior to this. The database model can be created using phpMyAdmin, which is an open-source database management system.
- A data dictionary will be created to collect descriptions of the data and objects in the database so that they can be viewed clearly in a single document.

User interface

Another objective of this project is to develop a front-end, web-based application that is responsive to both desktop and mobile devices. The website will provide an interface in which the stakeholders of this project can use to manipulate and interact with data in the backend.

Goals

- For the development of the web interface, wireframes will need to be created to design the website. High fidelity wireframes will be created using the Marvel Prototyping website. This website allows the creation of professional-looking wireframe designs.
- A professional-looking, fully functional website needs to be created for this project. The website will allow users to interact with the backend database. The website will be created using HTML, CSS and PHP. CSS stylesheets will be evaluated, tested and reviewed to find a design for the website that is intuitive and aesthetic.
- The website will need to be integrated into the backend database, so that is able to retrieve the relevant stored information and display it to the user on the webpage. The website will also need to store user input and send that back to the database.

1.3 - Method & Planning

The project methodology describes the process that will be used to manage the required tasks that need completing for the successful creation of the project. This project will follow the Agile Project Management Methodology. The Agile methodology breaks down projects into small pieces of work that periodically set for completion. The types of tasks completed will be designing, coding and integration. These periodic sessions are called sprints. The purpose of this methodology is to focus on the overall outcome and have minimal planning when possible.

The Agile Methodology is very adaptable. It works in incremental development cycles and therefore allows for the project deliverables to be adjusted when necessary. Thereout the development of this project, there will likely be changes to the project deliverables. It may be necessary to go back and forth with specific tasks, and an Agile development method is well suited to changes in a project.

When looking at product development planning structures, the Waterfall development methodology needs consideration. The waterfall method breaks down project tasks into sequential deliverables. Each phase relies totally on the completion of deliverables from the previous stage. If the previous deliverable is incomplete, it would not be possible to move onto the next deliverable.

The Waterfall method bases its philosophy on following a set of steps that keep people moving forward. The methodology does not leave much room for unexpected changes. Changes and revisions are likely to happen throughout the development of this project. If the project was using this method and the steps followed to the end of the project, but then an unexpected situation was to happen that required something to be changed, this would be difficult. A change to the initial

deliverables of the project could make the previous work completed redundant, which can cause delays in the overall completion of project deliverables and the entire project as a whole.
This development of this project can be broadly split into the following phases:

- Requirements Gathering
- Database Design & Development
- Website Design & Development
- Documentation

The completion of the three project phases is scheduled for completion in the order that they are listed above. However, it could be necessary to work on multiple stages simultaneously. It may be necessary to iterate back and forth between the stages.

- Stage 1 will need user requirements to be gathered from the client.
- Stage 2 will require an ERD design of the database. Once that is complete, the building of the database will start.
- Stage 3 requires the creation of wireframes for the website, and then the actual website needs to be created. Integration of the database to the website will also need to happen.
- Stage 4 is about the documentation that is required for this project; however, this will be an ongoing phase that will happen regularly at each of the three other stages.

1.4 - Gantt Chart

A Gantt Chart is a bar graph visualisation that shows the intended schedule of a project. It has been used in this project to break down each of the main project components into sections. Each section has been given an estimated completion time in the Gantt Chart. For the successful build of the project, each of sections in the Gantt Chart will need to be completed, in the order that the chart has listed their completion time.

Figure 1.1 - Gantt Chart

1.5 - Ethics

1.5.1 - GDPR

The PDMS will be storing information about students and supervisors, and therefore it is essential to follow GDPR. GDPR takes privacy law to account for more recent technical developments and how people use data. It increases restrictions on what organisations can do with user data, and it extends the rights of individuals to access and control data about them. If a user requests for their data to be removed from the system, by law it would need to be removed.

Not following the GDPR could result in penalties and fines. The maximum fine under the GDPR is up to 4% of annual global turnover or €20 million (whichever is greater). Therefore, data must be stored safely and securely in the PDMS so that no unauthorised third parties can gain access. This project will mitigate the risk of breaching the GDPR because the PDMS will be using fictitious open-source datasets which are available on the internet and accessible to anyone. Some data, such as the project deliverables for specific students, will need to be manually entered during the development of the system. This project will not be putting anyone's real personal information at risk because the datasets used will be fictitious and/or open source.

1.5.2 - Liability

If the PDMS were to malfunction and not correctly record a project deliverable, this could mean that a student forgets about an important task that needs completing. In a worst-case scenario, the incomplete task may cause that student to fail their project, and the PDMS would be mostly responsible for this. The affected student may then wish to sue the owner(s) of the system, which if the lawsuit is successful, would result in a hefty fine. To avoid this, when developing the system, it is essential that the core functionality of creating project deliverables on the webpage and having the information sent back and stored securely in a database works correctly.

2 - Literature Review

2.1 - Introduction

During a project, the techniques used to manage various people and tasks involved must be heavily considered. Excellent project management can increase the speed that a project is completed and the quality of the solution too. Poor management of a project can have the opposite effects meaning a project can take longer than anticipated, and the result of the project could fail to meet expectations (Bloch, 2020).

Project management has been scientifically researched over the years, and many practical strategies have been evaluated to discover effective ways that projects can be efficiently delivered. Different types of projects require different styles of project management; therefore, it is vital to pick the appropriate management style for specific projects. A study that recorded several IT projects found that 66% of software projects overrun expected costs, and 33% of software projects take longer than expected to complete (Bloch, 2020). Over-budget or over-due projects can have severe consequences to businesses, with extreme scenarios resulting in job-losses for employees, or bankruptcy for companies.

It is for this reason that a tactical and logical approach to software development is vital. In this chapter, two popular project management methodologies used by real-world software development teams will be examined so that the benefits of different methods can be evaluated. After evaluating the two popular project management techniques written in this chapter, developers and project managers will be able to make a logical and educated decision about which project management methodology is best suited for their specific project.

2.2 - Agile Project Management

The Agile project methodology breaks projects into smaller deliverable components. These components are usually referred to as ‘Sprints’. Sprints in Agile project management will have a series of tasks relating to the overall completion of a project. For example, if a project was to build a fully functioning website, a deliverable in one sprint could be to create the landing page of the website. Iterations in Agile project management are usually referred to as ‘Scrums' (Rouse, 2020).

Sprints are usually around two to four weeks long. At the beginning of each sprint, the team meets to discuss the upcoming project deliverables. After a Sprint, the project team usually meets to have a review of the previous deliverables (which should ideally be completed). Having sprints that usually only lasts for a few weeks allows for teams to demonstrate that the tasks completed in these segments are successful (at the end of a Sprint) and, if they are not, the team can then focus on fixing these issues quickly (Rouse, 2020).

As a result of reviewing project segments regularly, the chances of large-scale failures are drastically reduced because there is an improvement made to the system in each iteration (Scrum) throughout the project lifecycle (Rouse, 2020). In other project management methodologies (Waterfall), some large-scale errors may not be discovered until the end of the project, and by this time, it could be too late to make the required changes.

An Agile project manager is responsible for establishing clear roles and responsibilities to ensure proper engagement and accountability for specific areas of the project. It is also essential that an Agile project manager understands the effects of the interactions among a project's various parts and directs the people working on the project in the direction of continuous learning and adaptation (Augustine, 2020).

Dealing with unpredictable organisational environments is a challenge for managers of any software development project. Numerous real-world development efforts are likely to be handled in uncertain environments, as organisations adapt to changing technology, markets, and social conditions (Augustine, Payne, Sencindiver and Woodcock, 2020). With this in mind, requirements for systems must be able to change swiftly.

The Agile project management methodology allows for requirements to be evaluated and changed continually (more often than not) when necessary. Ostensibly minor changes can produce unanticipated effects in Agile projects. Unforeseen effects can happen because as systems are developed, their components are more likely to be more interdependent. As components are interdependent (Augustine, Payne, Sencindiver and Woodcock, 2020), making a change to one component can cause a domino effect and unintentionally alter other components. Project management approaches based on conventional linear development methodologies such as Waterfall are mismatched with such dynamic systems (Agile) in many circumstances, especially software development. This mismatch is in comparison to Agile is apparent when attempting to make iterative changes throughout the development cycle in the Waterfall methodology (Augustine, Payne, Sencindiver and Woodcock, 2020).

2.3 - The Waterfall Methodology

The Waterfall methodology is a project management approach where a project is completed in separate stages and moved step-by-step toward fulfilment. In the Waterfall methodology, an extensive plan is created upfront and then completed sequentially with each step relying on the completion of the previous one. This method assumes that there will not be any changes required in the project. If changes need to be made throughout the project, this method is not befitted to implement those changes (Bogdan-Alexandru, Andrei-Cosmin, Sorin-Catalin and Costin-Anton, 2020).

One person did not invent the waterfall methodology. Instead, it was named once people realised that there are other, more effective ways to manage software projects such as Agile project management. The Waterfall model is the earliest applied software development strategy, mirroring the management techniques that were used in different industries (Bogdan-Alexandru, Andrei-Cosmin, Sorin-Catalin and Costin-Anton, 2020). This strategy allows for a project to be divided into several fixed development phases. Each phase requiring the work from the previous stage to be completed:

The type of activity groups tasks on waterfall projects, and each project follows the same phases (The waterfall method, 2020):

- Requirements - Analyse the systems needs and document what software needs to do, think of functional and non-functional requirements.
- Design - Choose the technology that will be used for development, create diagrams and plan software architecture.
- Coding - Think about how to solve problems and write code to implement the solution.
- Testing - Make sure the code works as expected and does not break anything in the system.
- Operations - Deploy the system to a production environment so that the intended users can use it.

Waterfall was the first software development methodology. It was inherited from the manufacturing and construction industry where iterations are complicated and often not possible (after you have built a tower or a bridge you cannot go back to "improve" the foundation). However, because software development is predisposed to frequent change, the waterfall method is usually not the best solution in modern software development projects. Waterfall is often mentioned alongside Agile and stands in contrast to it. The main difference between the methodologies is that Waterfall does not react well to frequent adjustments, which is why it gets an inferior reputation in the software development community compared to Agile, where frequent changes are common and usually feasible (Olic, 2020).

The Waterfall method assumes that once the initial requirements are set, and every goal has been cleared of any ambiguity, there is an unobstructed metaphorical path that the development team must pursue to until the project is completed. However, in most real-life situations, this is not true, as customers/clients/stakeholders can alter their opinion towards different features of the project, which can mean that the project phases will have to be re-evaluated, frequently. This brings additional costs and time spent on different parts of the project, which could lower the overall customer/client/stakeholder satisfaction. This is the biggest flaw of the Waterfall model,

but it does not mean, this strategy should never be applied in a specific circumstance (Bogdan-Alexandru, Andrei-Cosmin, Sorin-Catalin and Costin-Anton, 2020).

An example of when the Waterfall Method could be used is when working on small projects with fixed deadlines, budget, and scope. In projects like this, the Waterfall model could help a team organise more effectively, as every member would have comprehensive documentation from the beginning and the knowledge lost if a member leaves the project is minimised. Time can be distributed optimally since every phase has a fixed period of time allocated for completion. However, in the eventuality that one task is late, all tasks are likely to be late. In the end, the product will have better cohesion since the design has been completed from the first phase, and everything has been taken into account. This makes for a more straightforward, streamlined process for the team, at the expense of any flexibility regarding future changes and additions (Bogdan-Alexandru, Andrei-Cosmin, Sorin-Catalin and Costin-Anton, 2020).

2.4 - Conclusion

To summarise this chapter, it is clear that for a project that involves software development, the Agile project methodology is often superior when compared to the classic waterfall method. The reason the Agile approach is more suited to software development is that it allows for project deliverables to be dynamic and adaptive. Having dynamic deliverables makes it that it easier for the development team to correct mistakes and apply changes. Mistakes are also easy to identify because the project team regularly meet to discuss how the project is going. In contrast, the waterfall method does not allow for dynamic changes to project requirements & deliverables; this is because the whole project development cycle is planned out beforehand and each stage of the development cycle is dependent on the previous phase being completed (Franklin, 2020)

3 - Requirements Chapter

3.1 - Introduction

The requirements analysis stage of a project is essential because it helps to identify what the aims are for the project and helps to determine if the project is feasible. The purpose of this chapter is to examine the necessary requirements needed for this project and also to examine the various stakeholders to see how this new system will positively assist them. Functional requirements precisely define what the system should do, and non-functional requirements define how the system should operate. In order to identify requirements, stakeholders of the project have been interviewed to find out what their specific needs are concerning the management of final-year university projects. The interviews were done to have an accurate understanding of what features the stakeholders would like to see in the solution. In addition to discussing both functional and non-functional requirements, this chapter will examine the specific stakeholders and how their requirements have been prioritised based on importance. This chapter also includes a use case analysis and discusses the potential volume of data that will be stored in the systems database.

3.2 - Stakeholder Analysis

This project has specific stakeholders that will have some interest or involvement in the end solution. This section will examine specific stakeholders to clarify the project's impact on them. This section will also examine the distinct roles of the stakeholders in relation to the project's development and the required level of engagement needed to satisfy their needs. Additionally, this section will review the current problems that are faced by specific stakeholders and what the benefits are for solving these. For this project, research was carried out to find who the specific stakeholders would be. The client, who is one of the main stakeholders, was asked to give his opinions regarding this. After discussing possible stakeholders, four main stakeholders were identified. Once the stakeholders were identified, ways of engaging the specific stakeholder categories were considered based on their overall influence on the project.

3.2.1 - The Stakeholders

One of the stakeholders of this project is Dr Graeme Jones, lecturer, final-year project supervisor at Kingston University and the client of this project. Every year, Graeme helps to manage the final-year projects of several students in the computing school of the university. Graeme would like a website that can be used to manage the projects of his students and keep track of the individual project deliverables that are agreed between himself and individual students. Graeme would like to make sure that project deliverables are completed on time and to a good standard, so it is clear which students are on track to completing their projects.

Other stakeholders of this project would be final-year university students who are completing final-year projects/dissertations and also their project supervisors. The final stakeholders

identified for this project are the parents of students. Parents would most likely want to ensure their children achieve high grades. Because of this, they will be interested in a project deliverables management system that helps their children to organise their projects.

3.2.2 - Current Stakeholder Problems

Currently, students face challenges when trying to keep up to date with the regular completion of their project deliverables to a high standard. Supervisors also face the problem of managing all of the deliverables that due to be completed by students. The PDMS addresses these problems by providing both students and supervisors with a centralised location where projects and their specific deliverables can be managed.

3.2.3 - Benefits of Solving the Problems

Students

The students would benefit from this system because they would have a centralised location where they can manage their final-year project and all of the deliverables that need to be completed. The ability to see tasks for a project along with receiving reminders when a task is near its due date would help to reduce the recurrence of forgetting that something needs to be completed.

The Client & Supervisors

Supervisors would benefit from this system for reasons similar to that of the students. The PDMS would provide a centralised location in which supervisors could manage the individual projects of their students. It would also provide supervisors with the ability to manage specific deliverables for these projects. It would provide an easy way to communicate with students about their final-year projects too.

Parents

Parents would benefit from this system because they would know that their children will be competently managing their final-year projects by keeping up to date with the required tasks.

3.2.4 - Stakeholder Analysis Table

Contact was made to the stakeholders stated below so that **table 3.1** could be populated accurately. Stakeholders play an integral role in the project and meeting their needs and expectations is vital regarding the success and outcome of this project. The opinions of all stated stakeholders will be considered, and the influence level of the stakeholder regarded in respects to a decision being made about the opinion. Those with significant influence are more likely to have their opinions implemented than those will a lower influence. If a stakeholder with weak influence had a conflicting opinion that a stakeholder with strong influence rating, it is likely that the opinion of the strong influence stakeholder will be prioritised.

Stakeholder	Impact of the project	Influence on the project	What type of engagement will be required?	What is important to the stakeholder?	How could the stakeholder contribute to the project?
Client	High	High	Keep satisfied and provide weekly updates about the project.	Creating a functioning system that will aid the management of organising projects for specific students.	Provide clear information about what he is expecting the system to do and be available to answer questions relating to the project.
Students	High	High	Keep satisfied and provide monthly updates about the project.	Building a system that allows the creation and management of tasks relating to projects.	Provide information of what would be important features of the system from his perspective
Supervisors	High	High	Keep satisfied and provide monthly updates about the project.	Creating a functioning system that will aid the management of organising projects for specific students.	Provide information of what would be important features of the system from his perspective
Parents of students	Low	Low	Keep satisfied and provide and update about the project once completed.	Making sure their children complete their final-year university projects to the best of their abilities.	Contribute their ideas about what they think would be useful for their children so that they stay on track.

Table 3.1 - Stakeholder Analysis Table

3.2.5 - Conclusion

The opinions of all stated stakeholders will be considered, and the influence level of the stakeholder regarded in respects to a decision being made about the opinion. Those with significant influence are more likely to have their opinions implemented than those will a lower influence. If a stakeholder with weak influence had a conflicting opinion that a stakeholder with strong influence rating, it is likely that the opinion of the strong influence stakeholder will be prioritised.

3.3 – SWOT Analysis

SWOT analysis is a technique that is used to identify the strengths, weaknesses, opportunities and threats for a business or project. The SWOT analysis has been used in this project to determine whether the proposed idea of the project was worthwhile and would be useful in the real world. After reviewing the information in the SWOT analysis table, it was clear to the developers that the project was indeed worthwhile. To combat the weakness that is stated in the SWOT table, in the future, the system could expand its functionality to focus not just on university projects, but all types of projects. It is difficult to combat the threat that is stated in the SWOT table, this is because some people may simply prefer to sue other systems, there is only so much that can be done to change people's preferences, however, after the system has been out for a while and grows in popularity, more users are likely to be drawn towards the PDMS.

Strengths	**Weaknesses**
The system will allow students to keep up to date with their project deliverables so that they can complete the project on time for its submission. The system will allow supervisors to keep track of the student's project work so they can manage each project efficiently.	The system is only going to focus on the management of final-year university projects, whereas other systems will allow people to focus on any type of project.
Opportunities	**Threats**
The system is going to provide a centralised location for students and supervisors to manage final year projects and therefore increase the overall manageability of the project. Usually, project management software focuses on multiple projects; the PDMS is specifically just going to focus on final-year university projects.	Some people may prefer to use existing solutions like Canvas, Trello or other project management software. Some people may prefer to use paper-based systems rather than a digital system for the management of their final-year projects.

Table 3.2 - SWOT Analysis

3.4 - Use Case Analysis

3.4.1 - Introduction

A user story is a means used in Agile software development to obtain a description of software features from an end-user viewpoint. The user story describes the types of users, what they want, and why. A user story helps to create a simplified description of a requirement. The idea of a user story is to document how a project will give value to the user. It is then the development team's responsibility to develop the software to meet the requirements of the user story. In this project, the developer will collaborate closely with the client and stakeholders to clarify the details as the system develops.

3.4.2 - Student User Story

"As a student, I would like to be able to manage my final-year project efficiently. Specifically, I would like to be able to create deliverable tasks within a project so that I can see what tasks I need to complete and when they are due. I would like to have the additional functionality of changing the due dates for tasks if circumstances were to change. I want to be able to upload images to the system so that my supervisor can see evidence of my completed work. I also want to receive emails reminding me when I have task deliverables due."

3.4.3 - Supervisor User Story

"As a supervisor, I would like to be able to monitor the progress of my students regarding their final-year projects. I want to see which deliverables they have set and also create deliverables for students to complete. I want to be able to see evidence of completed student work, whether that be from an image or an uploaded document. I want the system to be useable on both desktop and mobile devices. I also would like to be able to see a dashboard view of all my students which at a glance lets me see who is on or off track."

3.4.4 - Use Case Diagram

A use case diagram is a representation of the user's interactions with the system that shows the association between the user and the different use cases in which the user would be involved. A use-case diagram provides a clear overview of the system functionality. The diagram works as a blueprint and is a straightforward graphical illustration of what the system should do. Use-case diagrams are a useful communication tool for stakeholders because the designs mimic the real-world user interactions and provide a view for stakeholders to comprehend how the system will operate (Wikipedia, 2020).

3.4.5 - Use Case Description Catalogue

The purpose of a use case description catalogue is to summaries the various use cases of the system and all of the actors associated. The catalogue will additionally show the complexity and priority of each use case.

ID	Name	Actor(s)	Complexity	Priority
UC1	Upload Evidence	Student	High	7
UC2	Submit Work	Student	High	7
UC3	Create Deliverables	Student & Supervisor	High	10
UC4	Edit Deliverables	Student & Supervisor	High	10
UC5	Delete Deliverables	Student & Supervisor	High	10
UC6	Check Deliverables	Student & Supervisor	High	10
UC7	Login/Logout	Student & Supervisor	High	8
UC8	View Grades	Student & Supervisor	Med	5
UC9	Submit Feedback	Supervisor	Med	6
UC10	View Statistics	Supervisor	Low	5
UC11	View Student	Supervisor	High	8
UC12	View Evidence	Supervisor	High	7

Table 3.3 - Use Case Description

3.4.6 - Example Use Case Catalogue

The purpose of a use case catalogue is to provide information about specific individual use cases. In the example table below, there is information about which actors will have this use case, a description of the use case, use case frequency and more. Reading the use case catalogue table will provide a clear understanding of individual use cases. A comprehensive set of all of the catalogue description tables can be found in the appendix of this document.

Use Case ID	UC1
Use Case Name	Upload Evidence
Description	Allows the student to upload evidence of completed work
Actor(s)	Student
Frequency of Use	Will vary, anywhere from daily to never
Triggers	Supervisor wants to see proof of work
Preconditions	Student has access to internet and domain
Flow of Events	Log in to system, click on project, select evidence, then upload evidence
Exception Flow	The site is down for maintenance

Figure 3.4 - Use Case Catalogue

3.4.7 - Conclusion

The idea of a use case analysis is to distinguish the requirements of a software system. The use case analysis is the foundation from which the system software will be built. From the figures shown above, it can be seen that both students and supervisor use types of the system need to have the ability to do a range of tasks on the system. The primary purpose of the use case analysis was to design the system from the end-user's perspective and communicate system functions from the user's expressions.

3.5 - MoSCoW - Requirements

The MoSCoW prioritisation method is a technique used in business analysis, project management, and software development. The purpose of this technique is to reach an understanding with stakeholders about the importance of each project requirement. The requirements for this project have been categorised into the MoSCoW categories of 'Must' have, 'Should' have, 'Could' have, and 'Won't' have.

Must

Student CRUD (create, read, update, delete) Functionality

- As a student I want to be able to create a new task.
- As a student I want to be able to create a subtask.
- As a student I would like to be able to view ongoing projects and tasks.
- As a student I want to be able edit tasks.

- As a student I want to be able delete tasks.
- As a student I want to be able set a task deadline.

Lecturer CRUD (create, read, update, delete) Functionality

- As a lecturer I would like to be able to add projects and tasks for students to complete.
- As a lecturer I would like to be able to view individual student projects and tasks and also see all the projects that I have set for specific modules.
- As a lecturer I would like to be able to edit project descriptions and due dates.
- As a lecturer I would like to be able to delete projects.
- As a lecturer I would like to be able to add comments to tasks and projects.

Should

Communication Functionality

- As a student I would like to be able to receive feedback on tasks and projects.
- As a student I would like to be able to add comments to tasks and projects.
- As a student I want to be able view the grade given for a project.

Could

Monitoring Functionality

- As a student I would like to be able to view different tasks based on outstanding, late or completed.
- As a student I would like to be able to see a dashboard displaying all of my modules. When I click on a module, I would like to see which projects I need to complete. If I click on the projects, I should be able to see which tasks and subtasks need to be completed.
- As a student I would like to be able to view the due date of a task/project.
- As a student I want to be able change the status of a tasks/projects from ongoing to completed.
- As a student I want to be able view full project description and details.
- As a student I would like to be able to receive an email the day before a project is due for submission.

Would

Monitoring Functionality

- As a lecturer I would like to be able to view different student projects based on outstanding, late or completed.
- As a lecturer I would like to be able to view the due date of a project.

Communication Functionality

- As a lecturer I would like to be able to add feedback to tasks and projects.
- As a lecturer I would like to be able to see comments on projects/tasks from students.
- As a lecturer I want to be able assign a grade to a project.

Decision Making

- As a lecturer I want to be able mark a task as a priority.

3.6 - Conclusion

The purpose of this chapter was to clearly define the requirements for a successful build of the system so that it can be appropriately designed and implemented by the developers. The stakeholders of this project were examined; this will allow the developer team to create user interfaces based on the requirements of the end-users (as described in the stakeholder analysis and user stories). The use case section shows the interactions of the system for each of the end-users; this helps the developers to see what pages need to be created for the system. The functional requirements that have been prioritised using MoSCoW, clearly show the developers what needs to be created and what the most important features are for the project. In order to identify requirements, the project's client was interviewed in the summer of 2019.

4 - Design

4.1 - Introduction

The purpose of this chapter is to demonstrate how the requirements in the previous chapter will be implemented into the full system from the frontend webpage to the backend database. Various tools have been used to visualise the design elements of the system and also to show the intended workflows. This chapter contains an ERD and data dictionary to show a detailed view of various entities and objects involved in the database of the system. The ERD shows how the tables in the system connect to each other. The data dictionary breaks down the ERD and shows what data types will be used for attributes, and also gives examples of what data in the system will look like for each table. This chapter also contains sketches and wireframes to show the intended design of the frontend website. Following that, there will be a style guide to make sure that the webpage has a clear and consistent style. This chapter also contains a sequence and activity diagram to show the intended workflows of the system.

4.2 – Data Model

4.2.1 - Entity-Relationship Diagram

An entity-relationship diagram (ERD) is a data modelling method that graphically presents an information system's objects and the relationships between those objects. An ERD is a conceptual and representational model of data used to design the entity framework infrastructure. Each table in the ERD contains a unique table identifier, also known as a primary key. The primary keys allows for the referencing of one table in another. When using a primary key as a reference in another table, this is known as a foreign key and the ERD in figure 4.1 shows both primary and foreign keys. In addition to containing primary keys, tables will generally contain several other attributes that related directly to the table in which they are stored.

In figure 4.1, the ERD represents the database tables that will be created to store relevant system information. The ERD shows the attributes in each table and their associations to one another. By referring to the ERD, the project developers can create a relational database that implements the information shown in the diagram. Figure 4.1 has gone through many iterations to optimise the design of the systems backend database. The diagram has been discussed with the client of this project several times to ensure that the solutions backend design is capably of meeting the requirements that have been stated in the requirements chapter on this report and agreed on with the client.

The TaskStatus table stores the possible states for a project task. A task status can be ongoing, completed or overdue. The TaskStatus table provides a centralised location for all possible statuses to be seen. The table links directly to the Tasks table because a task will need to use the Status foreign key from the TaskStatus table.

The +0.* located below the evidence object in figure 4.3 shows that a project **may** have evidence and that it should be possible for a project to have **many** pieces of evidence. The +1.1 show in figure 4.3 means that a piece of evidence **must** belong to a project and a piece of evidence can only belong to **one** project only.

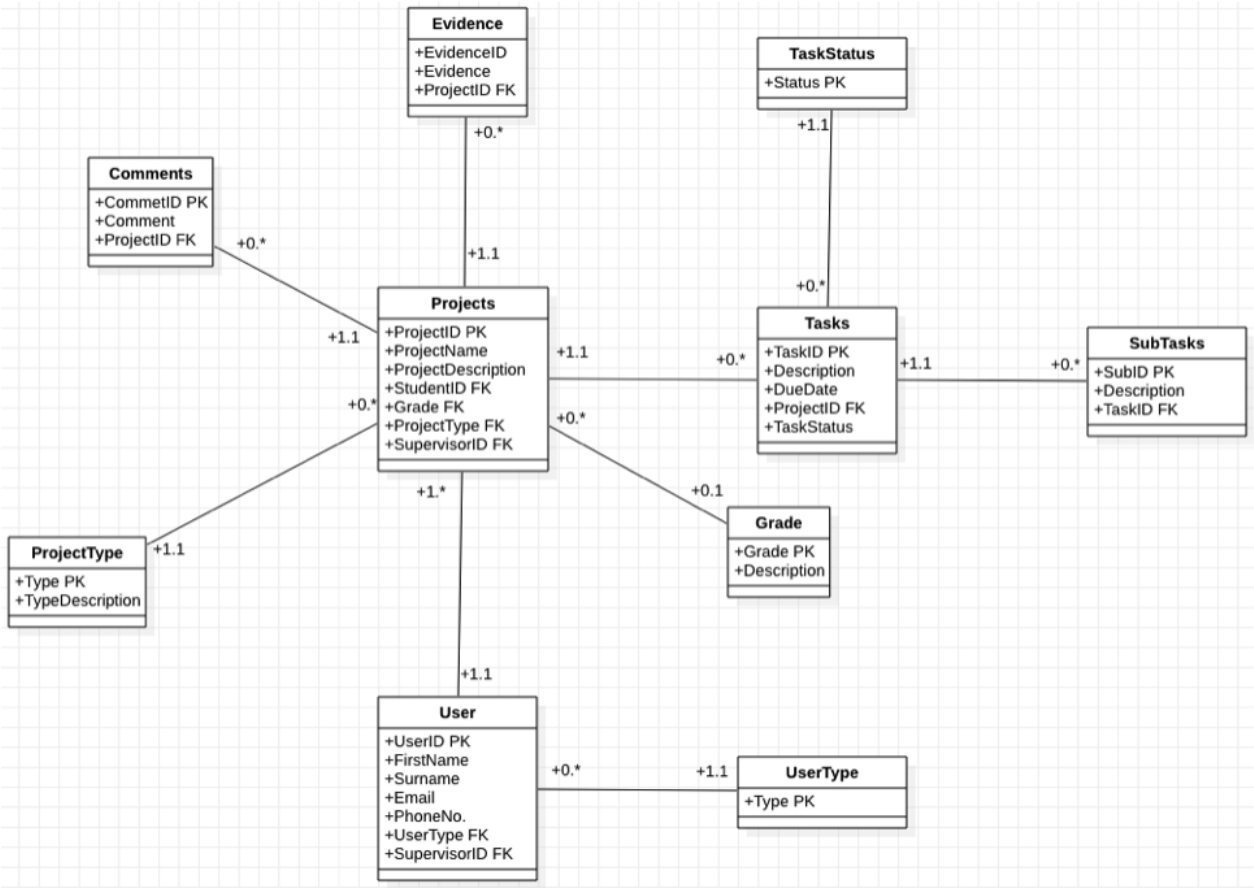

Figure 4.1 - Entity Relationship Diagram

4.2.2 - Data Dictionary

The purpose of a data dictionary is to provide comprehensive information regarding the contents of a database. For example, information such as the names of included variables, their data types, and examples of how specific data may look. A data dictionary provides a brief guide to understanding and using the data (Brandenburg, 2020). This chapter contains one example of a data dictionary table, for the full data dictionary, please refer to the appendix of this document.

Table 4.1 shows the data types that will be used for each attribute, the character lengths, a description of the attribute and also shows an example of how data in the database will look. The table also shows that the 'ProjectID' column is the primary key for this table and that StudentID, GradeID, ProjectType and SupervisorID are all foreign keys. To see the full data dictionary, please refer to the appendix of this document.

Field Name	Data Type	Character Length	Description	Example
ProjectID PK	Integer	5	Unique table identifier. Primary Key.	1
ProjectName	Varchar2	255	The title of the project	Management Website
ProjectDescription	Varchar2	255	The description of the project	To build a management description
StudentID FK	Integer	5	The unique identifier of the students table	10
GradeID FK	Varchar2	1	The unique identifier of the grades table	A
ProjectType FK	Varchar2	30	The unique identifier of the project type table	Build
SupervisorID FK	Integer	5	The unique identifier of the supervisor table	22

Table 4.1 - Data Dictionary

4.3 – User Interface

This section also contains sketches and wireframes to show the intended design of the website. Following that, there will be a style guide to make sure that the webpage has a clear and consistent style. This chapter also contains a sequence and activity diagram to show the intended workflows of the system.

4.3.1 - Sequence Diagram

A sequence diagram is a type of interaction diagram that describes how, and in what order a group of objects works together (Sequence diagram, 2020). The developers of the system will use this diagram to understand how the requirements for the system need to be implemented and how they will need to function. The diagram in figure 4.4 shows a user of the system logging into their account and performing a CRUD (create, read, update, deleted) operation. In the

diagram, it shows that when logging in to an account, the system needs to access the database to see if the login details are correct. If the details are correct, the user is shown their account information. If the details are incorrect, the system returns an error message. When the user is performing a create, read or update function, the system accesses the database, makes the changes (create or update) and then returns the information to the user on the system. If the user uses the delete function, the system accesses the database to removes the item that was deletion was requested for and then returns the deletion confirmation to the users.

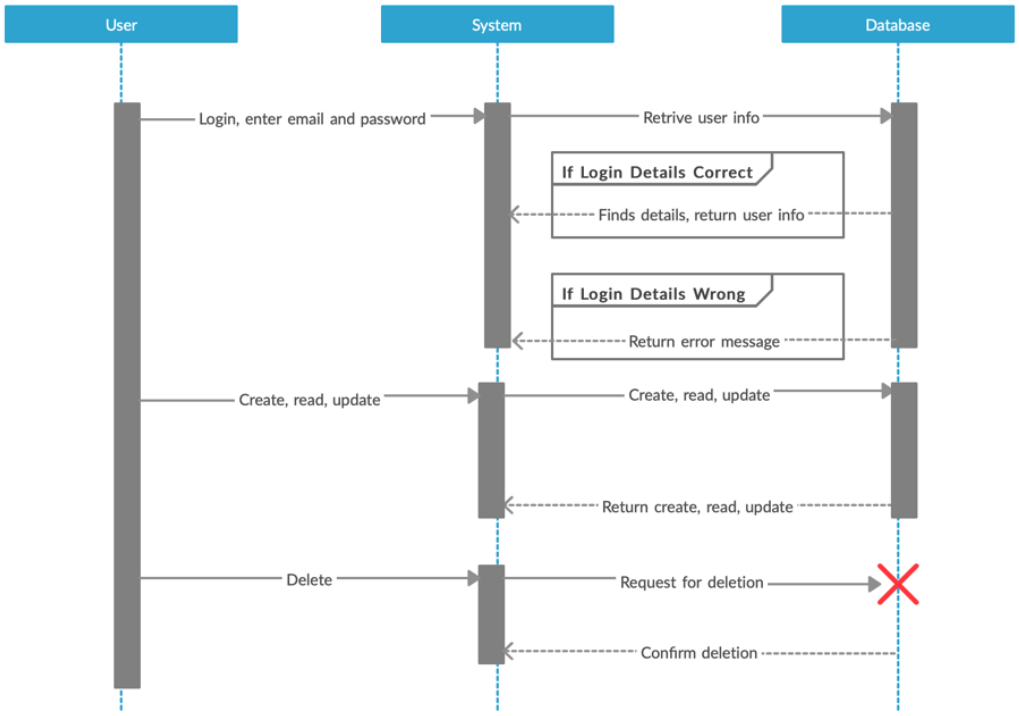

Figure 4.2 - Sequence Diagram

4.3.2 - Activity Diagram

Activity diagrams are graphical representations of workflows of following activities and actions with support for various choices from the user. The fundamental purpose of the activity diagram is to describe the dynamic behaviour of the system visually. The activity diagram is used to show message flow from one activity to another (Activity diagram, 2020). Figure 4.3 shows the possible sequences that a student user of the system can take. The student starts by logging into the website, from there, the student can then start to use the core functionalities of the system such as creating, editing & deleting projects & tasks. In an activity diagram, it is important to show all of the possible workflow options that a user can select on each page when navigating the system. The activity diagram ends with a user either creating, editing or deleting a task. From using the diagram below, the project developers can identify the exact workflow that they need to program into the system e.i. what webpages need to link to one another and also what buttons need to be on certain pages such as a ‘create’, ‘edit’ or ‘delete’ button.

4.3.3 - Sketches

Sketching is essential when designing a front-end webpage. Starting the design offline and on paper helps teams to visualise the project's potential early in the process. Sketching is one of the initial steps in the brainstorming process of any web design project. Visualising a project together from the start can help the team to get a precise idea of how the final design may look. Sketching is a method used to help clarify design ideas, provide meaningful feedback and keep the project progressing forward. Sketching can also cultivate agile development and change. The design team can try out ideas on paper and doing this allows the developers to go back to the code and figure out how to turn those ideas into practical solutions.

Figures 4.5 shows the initial sketched design of the projects page of the system. The idea for this page is to show the user comprehensive view of all their ongoing projects. Figure 4.6 is the edit project page which has the purpose of allowing a user to alter the name and description of a project. To see all of the initial system design sketches, please refer to the appendix of this document.

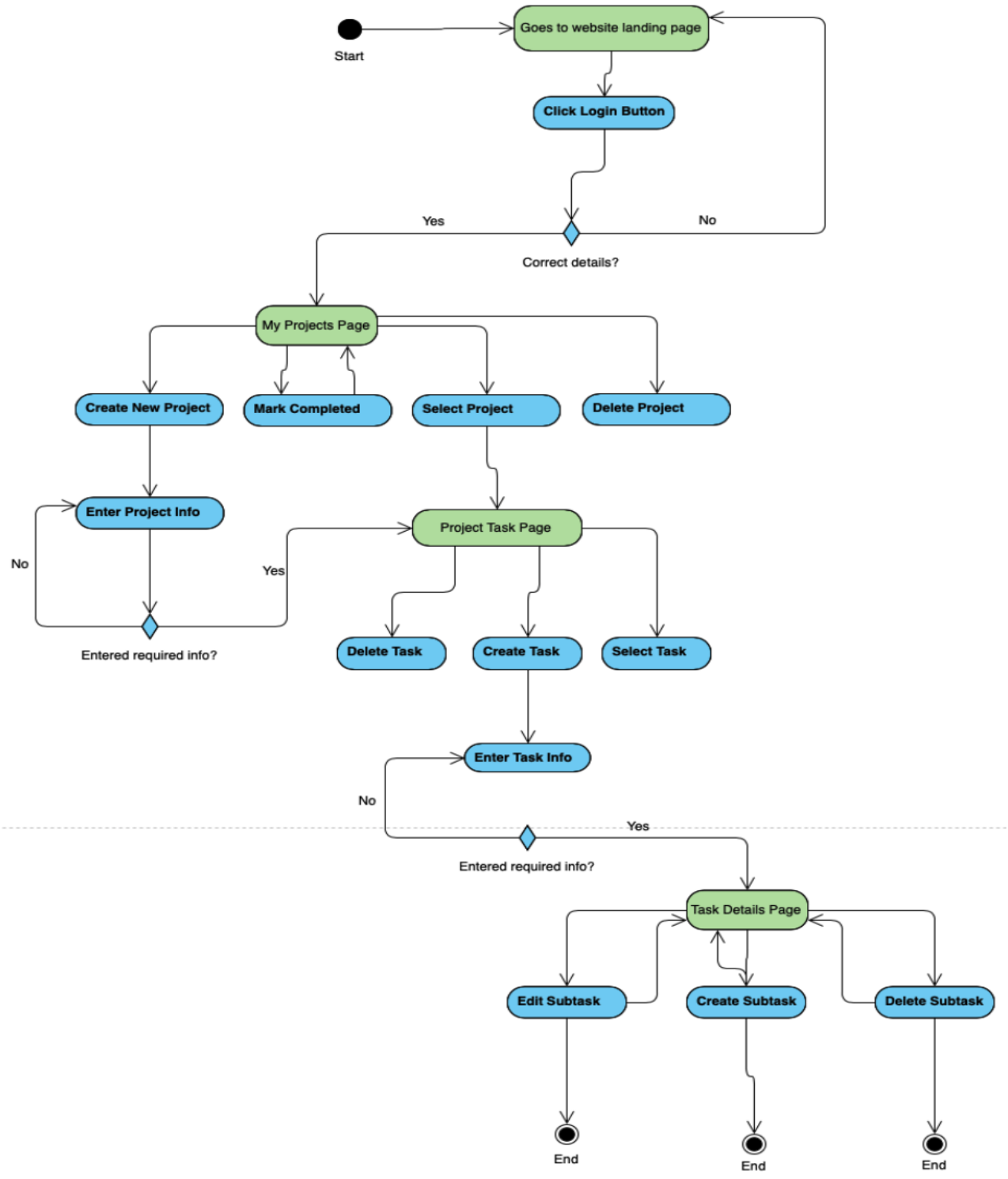

Figure 4.3 - Activity Diagram

4.3.4 - Mid-Fidelity Wireframes

Wireframes have been used to lay out the content and functionality of various pages of the system. The wireframes shown in figure 4.4 take into account the requirements that have been mentioned in chapter two of this document. It is essential to use wireframes early on in the development process to establish the basic structure of a page before final design and content is created. The wireframes provide a foundation from which the final solution can be built from. When creating wireframes, the systems designers can identify user interface/user experience design flaws; spotting design issues early in the design process is important so that the final solution is a problem free as possible.

Figure 4.4 shows the mid-fidelity wireframes from the Projects page of the system. On Projects page, users are able to view all of their ongoing projects in a card-like layout. Clicking on a particular card will expand the project and show the user the full project details, along with any tasks that have been created for that project. On this page there is also a create projects button which allows the user to create a new project to be added to the projects page. To see all wireframes, please refer to the appendix of this document.

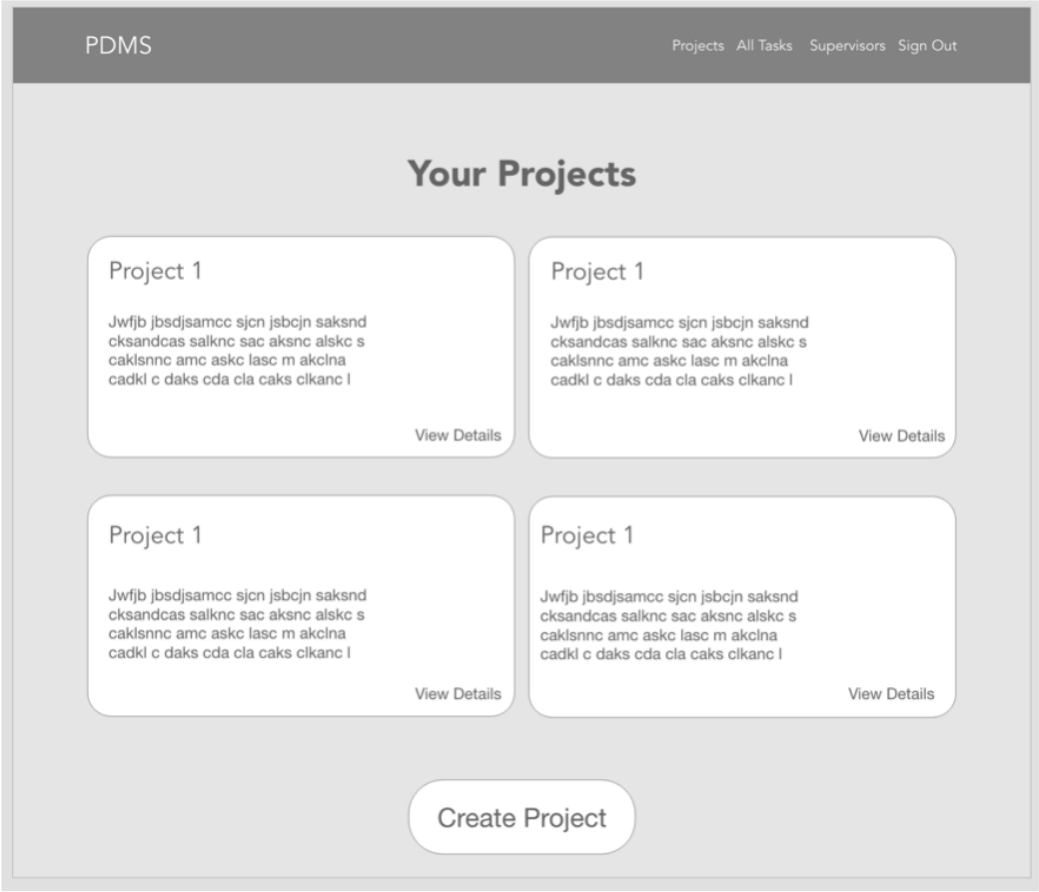

Figure 4.4 - Wireframes

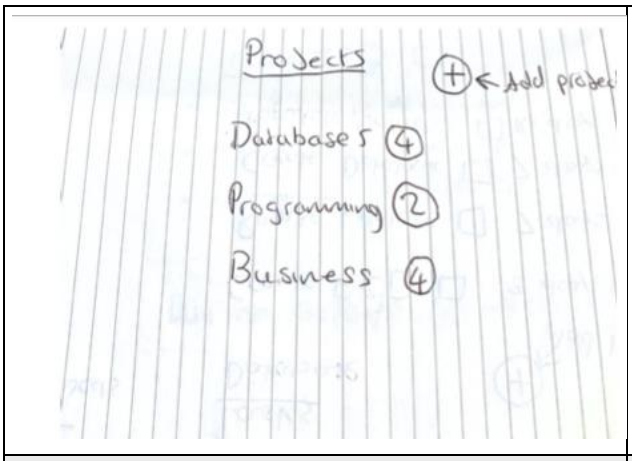

Figure 4.5 - Project Page

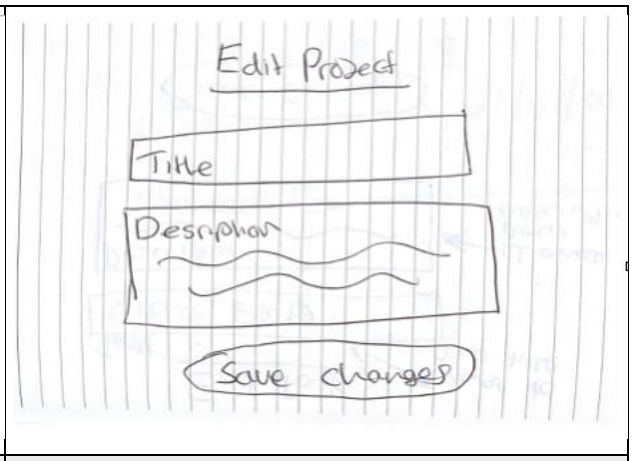

Figure 4.6 - Edit Project Page

4.4 - Conclusion

The design stage is an integral part of any project; it sets a framework for the aesthetic of the system. The ERD and data dictionary design shows the database developers how the systems backend data needs to be stored and managed. The sequence diagram shows what technical elements are needed to support the desired workflow, and the activity diagram helps the developers to understand what the workflow of the system should be. The design elements of this chapter were created upon the requirements listed in chapter 2 of this report. The sketches and wireframes, included in this chapter, helps the developers to understand the desired look of the website. The sketches and wireframes are initial ideas and could very well be changed in the future; the benefit of creating these is to provide a backbone for the user interface of the system to be built.

5 - Implementation

5.1 - Introduction

The implementation of this system is one of the most critical aspects of this project. It combines all of the previous steps from the requirements chapter and the design chapter. The implementation chapter is about turning the concepts of the project into a reality so that the system can be used in the real world.

5.2 - Coding Principles

Code Reuse

Code reuse is about reusing exciting software that has already been created to deliver software functionality. Code reuse has been done extensively by the project developer of the system. Several pages of the website require similar functionalities, meaning that once the function for one requirement has been created, it can be reused and slightly modified for another. To create all of the required functionality for the system, the developers looked at various source such as Stack Overflow, and YouTube. Some elements of code solutions were inspired from these sources, however, the developers customised the code to fit the specifics of the system being created (anuupadhyay, 2020).

KISS

It is inconvenient for developers to frequently debug, maintain or make changes in complex code. "**Keep It Simple, Stupid (KISS)**" states that most systems work optimally if they are kept simple rather than making complex. Therefore, when developing the solution, it should not be so complicated that takes a lot of time and effort to understand. If the code is simple, other developers will not face any problem understanding the code logic which makes it easier for the project to be developed further. In future, the project could be handed off to another development team, therefore, having code that is easily interoperable is vital. So always try to simplify your code using different approaches like breaking a complex problem into smaller chunks or taking out some unnecessary code you have written (anuupadhyay, 2020).

DRY

Duplication of data, logic or function in code makes code unnecessarily lengthy and also wastes time when it comes to maintaining, debugging or modify the code. If a small change is needed in the systems code, developers will need to do it at several places. The **"Don't Repeat Yourself (DRY)"** goal is to reduce the repetition of code. DRY states that a piece of code should be implemented in just one place in the source code. The opposite of the DRY principle is WET ("write everything twice" or "waste everyone's time") which breaks the DRY principle if you are

writing the same logic at several places. You can create a common function or abstract your code to avoid the repetition in your code (anuupadhyay, 2020).

YAGNI

The system could become larger and more complex if the developers are writing some code which might be needed in future but not at the moment. **"You Aren't Gonna Need It (YAGNI)"** principle states "do not implement something until it is necessary" because in most of the case, this piece of code is not going to be used again in the future. While implementing software, many developers think about the future possibility/functionality of the system and then write code for features which they don't need at present. Developers often add all the unnecessary class and functionality which maybe they never use in the future. Doing this will eventually end up in writing bloated code which means the project becomes complicated and difficult to maintain. In this project it was recommended to the developers to avoid this mistake to save a lot of time and effort (anuupadhyay, 2020).

5.3 - Technologies & Resources

The Project Deliverables Management Systems a web-based application. It has been created using current web technologies and is accessible on most web browsers such as Google Chrome, and Safari on both desktop and mobile devices. Several tools were used for the creation of the system.

StarUML

For the design of the database ERD, the project developers will be using StartUML. This is a great application to use when designing database models and will help show a visual representation of the database. The database design went through several iterations. Discussions were had between the client and the developers to ensure that the ERD was designed in a way that would enable to system to achieve the essential project requirements. Using StarUML, allowed for the developers to clearly translate ideas in their minds onto paper. In future projects, the developers may decide to use another UML creation tool such as draw.io. draw.io is a website that allows for UML designs to be saved on the cloud, whereas StarUML must be download to the user's machine.

Visual Studio Code

Visual Studio Code has a user-friendly environment, that is both intuitive and easy to use. Alternatively, Notepad++ could be used. This program was used to develop the code for the system. Initially, the project developers used the code editor named 'Atom', however, after using that application for a few weeks, a decision was made to use an alternative code editor, due to the slow and often faulty performance of Atom. Visual Studio Code provided the developers with

speedier performances and instances of the software glitching/crashing were next to none. In future projects, Visual Studio Code is likely to be the first-choice code editor for the developers.

phpMyAdmin

PhpMyAdmin is a free and open source administration tool for MySQL and MariaDB. As a portable web application written primarily in PHP, it has become one of the most popular MySQL administration tools, especially for web hosting services. PhpMyAdmin has been used to manage the database of the system. PhpMyAdmin allows for the creation and manipulation of database tables. It also allows for large datasets to be inserted. Through the phpMyAdmin interface, developers can test SQL code to test that it does what is expected before implementing it into the website.

Adobe XD

Adobe Experience Design was used to test different designs for the project deliverables management system. Mid-fidelity interfaces for both desktop and mobile devices were designed using this application. Adobe XD is design tool used by user experience/user interface designers to test out design ideas before they are translated into a functional application. The project developers experienced no issues when using Adobe XD and are likely to use the application again in future project designs.

XAMPP

The website files have been hosting using XAMPP. XAMPP is a free and open-source cross-platform web server solution stack package developed, consisting mainly of the Apache HTTP Server, MariaDB database, and interpreters for scripts written in the PHP and Perl programming languages. Storing files on XAMPP allows PHP code and database to be ran on the local machines of the project developers. (Wikipedia, 2020).

Mockaroo

Mockaroo is a website that automatically generates datasets with potentially thousands of rows of data. This website allows users to choose the the types of attributes that they want in their database tables, and then the system automatically generates rows of data with random entries such as first names, phone numbers etc.

5.4 - Scripting Languages

HTML

Hypertext Mark-up Language is the standard mark-up language for documents designed to be displayed in a web browser. It can be assisted by technologies such as Cascading Style Sheets and scripting languages such as JavaScript. HTML has been used to configure the content that the system displays.

CSS

Cascading Style Sheets is a language that can alter how a mark-up language, HTML, is presented through alterations of layout, size, font, colour and many other aesthetic variances. CSS has been used to style the content of the system. The design of the PDMS website was inspired by existing to do list applications, along with the wireframes that were created for the system. CSS was used to make the website responsive to devices of different screen sizes. The website was made with a 'mobile-first' ideology, meaning that the mobile interface was designed first. The mobile-first design strategy was used because most people (especially the target users of this project) will be using their phone to access most web-applications.

PHP

PHP will be used for communications to the database in the backend which with phpMyAdmin will manage the database through database/table creation, deletion, renaming, copying and inserting. It can also be used for general programming purposes too. PHP has been used to make connections to the information in the database and either send information back from the website or retrieve information to be displayed on the website.

JavaScript

JavaScript is a programming language that can add interactivity to a webpage and is considered one of the three essential technologies of website development along with HTML and CSS. Javascript has been used for the drop-down calendar form function of the system.

SQL

Structured Query Language is purpose-built for the management of data in relational database management systems and processing in relational data stream management system. This language has many functionalities, amongst them; inserting, editing and deleting data, creating, updating and deleting tables, displaying and comparing records.

5.5 - Data Generation

5.5.1 - Introduction

For the successful implementation of this project, data needed to be generated so that the tables in the database could be populated with information that resembled what would be stored in real life. When a webpage has lots of data displayed, it can look different compared to when there is very little data.

5.5.2 Purpose of Data

It is essential to insert lots of data in the system so the developers can test if the styling of the webpage holds together. Sometimes, designers will think of an initial design that looks good when there are minimal amounts of data. However, once the number of data increases, the initial

design does not keep its intended styling. During the creation of this project, the developers need to redesign the webpage. The reason for the redesign was due to the website styling changing after hundreds of rows of data were added. The design was updated to accommodate the new rows of data.

5.5.3 - How the Data was Generated

Data was inserted into the various tables in the database by a variety of methods. In this chapter, the methods used will be examined along with the pros and cons of the respected methods. The datasets used in this project are mainly supposed to simulate students, supervisors and components relating to projects in the system. Several rows of data have been inserted into each table. The data entered is supposed to simulate a scenario where the system has been out for several years, and there is a significant amount of data populating the tables.

5.5.4 - Mockaroo & Excel

Mockaroo is a website that automatically generates datasets with potentially thousands of rows of data. This website allows users to choose the types of attributes that they want in their database tables, and then the system automatically generates rows of data with random entries such as first names & phone numbers. Mockaroo was used to create rows of data for the Users table of the system. All rows were populated, and then an Excel file was created & downloaded from the website. After downloading the document, the Users file was opened in Excel. In the Users table, there is a foreign key named 'UserType'. UserType could not be generated using Mockaroo. Therefore, each user had to be given a UserType by manually manipulating the Excel document. Each user was given a 'type'. Then, the document was converted to a CSV file and uploaded directly into the Users table of the database.

Most tables in the system were populated by using a combination of Mockaroo & then manually manipulating the rows of data in Excel. These tables include Comments, Evidence, Users and Projects. The Tasks table was populated by using Graeme Jones' (project client) task deliverables spreadsheet. Graeme uses this spreadsheet with his final-year project students to help keep them on track with work. The spreadsheet contains over 100 tasks. Tasks were copied from Graeme's spreadsheet, placed onto a new Excel file and then uploaded to the tasks table in the database. Each project in the system was assigned roughly 50 tasks.

```php
<div class="cards-container">
    <?php foreach($projects as $project){ ?>
        <a href='tasks.php'>
        <div class="card">
            <div class="container">

            <form class="" action="insertproject.php" method="post">
                <h4><b><?php echo htmlspecialchars($project['ProjectName']) ?></b></h4>
                <p><?php echo htmlspecialchars($project['ProjectDescription']) ?></p>
            </div>
         </a>
            <button>
                <a href="createproject2.php?edit=<?php echo $project['ProjectID']; ?>">Edit</a>
            </button>
            <button>
                <a href="deleteproject.php?delete=<?php echo $project['ProjectID']; ?>">Delete</a>
            </button>
        </form>

        </div>
    <?php } ?>
</div>
```

Figure 5.1 - HMTL/PHP code to show the 'cards-container' layout.

```css
/* Add some padding inside the card container */
.cards-container {
  padding: 10px;
  margin: 20px;
  text-decoration: none;
}

.cards-container, h4, a, p, div {
  text-decoration: none;
}


/* Wraps cards */

.cards-container {
    display: flex;
    flex-flow: row wrap;
}
```

Figure 5.2 - CSS code to show the 'cards-container' customisations.

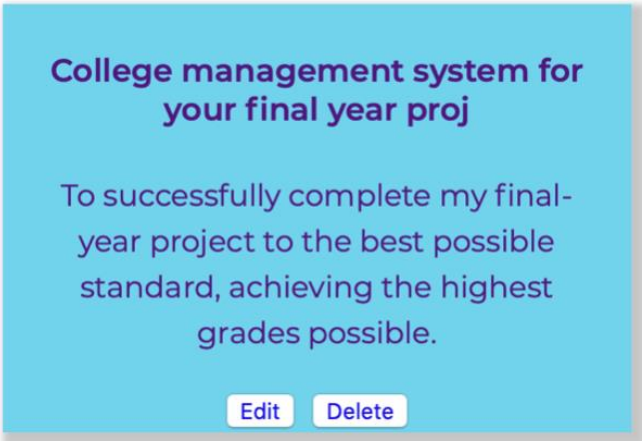

Figure 5.3 - Website frontend showing the result of the card layout in the. HTML & CSS document.

```php
<?php

// connect to SQLiteDatabase
$conn = mysqli_connect('localhost','Kevon', 'Jamess1.2?', 'PDMS');

// check connection
if(!$conn){
  echo 'Connection error: ' . mysqli_connect_error();
}

//  Queries
$sql11 = 'SELECT * FROM Projects INNER JOIN Students ON Projects.UserID = Students.UserID';
$sql = 'SELECT * FROM Projects WHERE UserID = 2179';

// make query and get results
$result = mysqli_query($conn, $sql);

// fetch resulting rows as ab array
$projects = mysqli_fetch_all($result, MYSQLI_ASSOC);
```

Figure 5.4 - PHP code connecting to database and executing an SQL

5.6 - Functions Created

The main functions implemented to the systems were based around CRUD functionality. In computer programming, create, read, update, and delete (CRUD) are the four basic functions of persistent storage. Alternate words are sometimes used when defining the four basic functions of CRUD, such as retrieve instead of read, modify instead of update, or destroy instead of delete. By applying CRUD functionality to the system, the users will be able to properly manipulate relevant content on the website. Some form CRUD functionality is vital for most pages of the system, with some pages requiring multiple CRUD actions.

Create

One of the main functional requirements for the system was for students to be able to create a project. To do this, the user would need to enter their project's name, description, project type and also assign their desired supervisor. This critical functional requirement has been successfully implemented to the system, using a combination of HTML, CSS, SQL & PHP code. It was also a requirement for student users to be able to create tasks relating to a project, and subtasks relating to a task. Both of these requirements have been successfully implanted and work as intended on the website. The 'create project' function was created first, then the 'create tasks/subtasks' function. After successfully creating the initial create function, the code could

then be used for similar requirements. To learn how to implement the create function, the developers of the project looked at online martial from sources such as YouTube.com and W3Schools.com.

Read

Being able to read information from the database on the frontend website is a must-have functional requirement of the system. To do this, PHP & HTML were used. The first instance of this function being created was on the 'Projects' page of the website. The website needs to return all projects that a student has ongoing. To do this, an SQL 'SELECT' query was impeded into PHP a variable and the results stored in an array. The items in the PHP array were then printed to the screen through a mix of HTML & PHP. The developers of the system were able to implement this function swiftly. Once this function was completed for the projects page, the code was then resumed for other pages of the website such as 'Tasks', 'Subtasks', 'Comments' & 'Evidence'. Each of these pages require the Read function of CRUD to be implemented.

```
<div class="cards-container">
    <?php foreach($projects as $project){ ?>
        <a href='tasks.php'>
        <div class="card">
            <div class="container">

            <form class="" action="insertproject.php" method="post">
                <h4><b><?php echo htmlspecialchars($project['ProjectName']) ?></b></h4>
                <p><?php echo htmlspecialchars($project['ProjectDescription']) ?></p>
            </div>
        </a>
            <button>
                <a href="createproject2.php?edit=<?php echo $project['ProjectID']; ?>">Edit</a>
            </button>
            <button>
                <a href="deleteproject.php?delete=<?php echo $project['ProjectID']; ?>">Delete</a>
            </button>
        </form>
```

Figure 5.5 - shows code that is used to display all of the projects from the database to the webpage. To PHP uses a 'for each' loop to iterate through all of the projects stored in the database, and then displays the project name & description of the project in a card format.

```php
//  Queries
$sql11 = 'SELECT * FROM Projects INNER JOIN Students ON Projects.UserID = Students.UserID';
$sql = 'SELECT * FROM Projects WHERE UserID = 2179';

// make query and get results
$result = mysqli_query($conn, $sql);

// fetch resulting rows as ab array
$projects = mysqli_fetch_all($result, MYSQLI_ASSOC);
```

Figure 5.6 - the query to retrieve the projects in the database is ran using the code in this figure. The results of the query are then stored in a PHP array named ‘$projects’. The items in the $projects array are called on in figure x so that they can be individually shown in on the webpage.

Edit

This was the most difficult feature for the systems developers to implement. The development team encountered many errors when attempting to successfully implement this, however, eventually they were able to complete this key functionality. Users of the system should be able to edit various components of the system such as projects, tasks etc. To do this, PHP code was used because it allows developers to connect to the backend database. Hours of research and learning was done be the developer so that they could successfully apply this feature to the website. The first page that this feature was applied to was the ‘Projects’ page. After the implementation on this page was successful, the code could be reused on other pages that required similar functionality.

One issue that the development team encountered regarding this function was that variables in PHP are not immediately available outside of the page where they were initially created and assigned a value. Variables are needed for the edit function of the system to store and return the values in the database that the user would like to edit. Initially, the development team had the PHP code responsible for editing the database saved on a separate page from the HTML frontend page. To resolve this, the PHP variables were created and initialised in the same document to that of the frontend HTML page. After making this change, the edit functionality was working, and the development team knew how to create this function for future pages of the website.

The PHP code could have been kept separate from the HTML code if the developers choose to use the ‘session’ function provided by PHP, however, this option was decided against in a attempt to reduce the complexity of the code. In future projects when more time is available, the session PHP function might be used by the developers.

```php
if (isset($_GET['edit'])){
    $id = $_GET['edit'];
    $update = true;
    $result = $con->query("SELECT * FROM Projects WHERE ProjectID=$id") or die($con->error());
    $project = $result->fetch_array();
    $name = $project['ProjectName'];
    $description = $project['ProjectDescription'];
}

if (isset($_POST['update'])){
  $id = $_POST['id'];
  $name = $_POST['ProjectName'];
  $description = $_POST['ProjectDescription'];

  $con->query("UPDATE Projects SET ProjectName='$name', ProjectDescription='$description' WHERE ProjectID=$id") or die($con->error);
  header("location: projects.php");
}
```

Figure 5.7 - 'Edit' PHP Code - The first IF statement is responsible for retrieving the values that the user would like to edit. The second IF statement is responsible for updating the database record with the updated value that the user entered.

Delete

The delete function is necessary because users of the system should be able to remove previously created projects and tasks from the system. The delete functionality took the developers a significant amount of time to implement because they had no prior knowledge of coding this particular functionality. Extensive research was done on the Internet, the main source being YouTube.com. Eventually, the developers we successfully able to implement this feature to the website and it can now be used on various pages of the website. To implement this feature, PHP was used because it allows the webpage to connect and interact with the database.

5.7 - Discussion on Coding Issues

Coding Languages

The developers of the system did not have much experience using PHP prior to this project, therefore lots of learning needed to be done to implement the core functionalities of the website. PHP was used because it was the most suitable coding language that could be used to create the features of the website. Some features of PHP were picked up quite easily by the developer, whereas others took more time to learn. The developers did have prior knowledge of HTML, CSS & SQL which were all needed to in conjunction to create many of the required functions of the system. Throughout the projects, the developers enhanced their knowledge of all of the previously listed programming languages, therefore in future projects, implementing similar features will not be as difficult and can be done at a faster rate.

Delete Function

The delete function took a several hours for the development team to implement. To successfully create this function, the developers needed to combine their HTML, SQL and PHP knowledge. A delete button was created using HTML. Upon clicking this button, the system would apply the PHP & SQL code needed to delete the item. To learn how to do this, the project developers looked at various resources on YouTube and Stack Overflow until they had sufficient knowledge to implement this code themselves.

```
<button>
    <a href="deleteproject.php?delete=<?php echo $project['ProjectID']; ?>">Delete</a>
</button>
```

Figure 5.8 - Delete button for the HTML page of the website.

```
if (isset($_GET['delete'])){
  $id = $_GET['ProjectID'];
  $con->query("DELETE FROM Projects WHERE ProjectID=$id") or die($con->error());

  $_SESSION['message'] = "Record has been deleted!";
  $_SESSION['msg_type'] = "Success!";

  header("location: projects.php");
}
```

Figure 5.9 - Clicking on the code shown in figure 1 will direct the system to this page. The 'if(isset(' code essentials means that if this button is clicked, then run the code. The code runs a query in the database to delete the project that the user has selected. The 'header' code at the bottom of the image tells the system to redirect to the projects page after this code has been successfully executed.

```
if (isset($_GET['delete'])){
  $id = $_GET['delete'];
  $con->query("DELETE FROM Projects WHERE ProjectID=$id") or die($con->error());

  $_SESSION['message'] = "Record has been deleted!";
  $_SESSION['msg_type'] = "Success!";

  header("location: projects.php");
}
```

Figure 5.10 - Shows code that was previously used by the developers, however, did not work because there was an error in the circled location. This caused an error because the variable stored in $_GET was 'ProjectID', whereas what needed to be stored was the actual ID number of a specific project. This was corrected and the delete function now works as intended.

ProjectType - Drop Down List

This feature took several hours for the developers of the system to implement. The drop down list needed to return information from the 'Project Type' backend database. To present the options in the in the database to the frontend webpage required some PHP code. The developers used a 'while loop' to iterate through the options in the ProjectType table of the database. The developers did not initially know how to create this functionality, therefore they researched how to add this feature online through sources such as StackOverflow.

```
<select class="project-type-div" name="ProjectTypeFK">
    <!-- <option selected disabled>Select Project Type</option> -->
    <?php while ($row1 = mysqli_fetch_array($result1)):;?>
        <option class="project-type-form"
        name="ProjectTypeFK"><?php echo $row1[0];?></option>
    <?php endwhile;?>
</select>
```

Figure 5.11 - Shows a form that returns of drop-down list for all of the different project types in the database.

Figure 5.12 - Shows the frontend webpage result after implementing the code shown in figure 5.11

Bootstrap & CSS

In earlier design iterations of the project, the developers were using bootstrap for the design of the system, however, the team could not fund a bootstrap design that they liked enough to use in the final system. Therefore, the developers made the decision not to use bootstrap. Instead the developers decided to use their existing CSS knowledge to manually create the design of the system themselves. To do this to a high standard, extra learning on CSS was required so that all of the ideas from the design team could be implemented into the live system. To learn additional information about CSS, the developer mainly used YouTube and W3schools as their main sources of guidance. The responsive design elements of the system were manually coded by the developers too so that the system would look good no matter what platform the enduser was

operating. Overall, the CSS design aspect of the project did not cause many issues and was done with minimal complications. Any complications that the team encountered were easily debugged and looking at exciting examples available on the internet.

Create Function

The create function took a several hours for the development team to implement. To successfully create this function, the developers needed to combine their HTML, SQL and PHP knowledge. A create button was created using HTML. Upon clicking this button, the system would apply the PHP & SQL code needed to create the item. This function was first implemented on the projects page of the system, the code was then reused on several other pages that had similar functionality. The function required the system to get information from the user (project name, description), and return that information to be stored in the database. To do this, forms were used to capture user input.

```
<!-- Form -->
<form class="project-form" action="insertproject.php" method="post">

    <div class="project-name-div">
        <label>Project Name</label>
        <input type="text" name="ProjectName" class="project-name"
        value="<?php echo $name; ?>" placeholder="What is the name of your project?" id="ProjectName">
    </div>

    <div class="project-description-dv">
        <label>Description</label>
        <input type="text" name="ProjectDescription" class="project-description"
        value="<?php echo $description; ?>" placeholder="What is your project about?" id="ProjectDescription">
    </div>

    <select class="project-type-div" name="ProjectTypeFK">
        <!-- <option selected disabled>Select Project Type</option> -->
        <?php while ($row1 = mysqli_fetch_array($result1)):;?>
            <option class="project-type-form"
            name="ProjectTypeFK"><?php echo $row1[0];?></option>
        <?php endwhile;?>
    </select>

    <select class="supervisor-type-div" name="SupervisorFK">
        <!-- <option selected disabled>Select Supervisor</option> -->
        <?php while ($row2 = mysqli_fetch_array($result2)):;?>
            <option class="supervisor-from"
            name="SupervisorFK"><?php echo $row2[0];?></option>
        <?php endwhile;?>
    </select>

    <input type="submit" name="insert" value="Create" class="cta-project">

</form>
```

Figure 5.13 - shows the code for the form that is displayed on the create project html page. Information from the form is then sent to the 'insertproject.php' file; this is done by using the 'method=push' function, located at the top of the figure.

```php
$ProjectName = $_POST['ProjectName'];
$ProjectDescription = $_POST['ProjectDescription'];
$ProjectTypeFK = $_POST['ProjectTypeFK'];
$SupervisorFK = $_POST['SupervisorFK'];

$sql = "INSERT INTO Projects (ProjectName, ProjectDescription, UserID, SupervisorID, ProjectTypeFK)
VALUES ('$ProjectName','$ProjectDescription','2179','$SupervisorFK','$ProjectTypeFK')";
```

Figure 5.14 - Information from the form in figure 5.13 would then be passed into the 'insertproject.php' page. This page would store the results in the form into variables, and then insert those variables into the required columns in the projects table.

```php
if(!mysqli_query($con, $sql))
{
  echo 'Not inserted';
}
else
{
  echo 'Inserted';
}

header("refresh:2; url=projects.php");
```

Figure 5.15 - after executing the code in figure 5.14, the user is shown a message saying 'inserted' or 'not inserted' to let them know whether the creation was successful or not. After the message has been displayed, the user is redirected back to the projects.php page after 2 seconds.

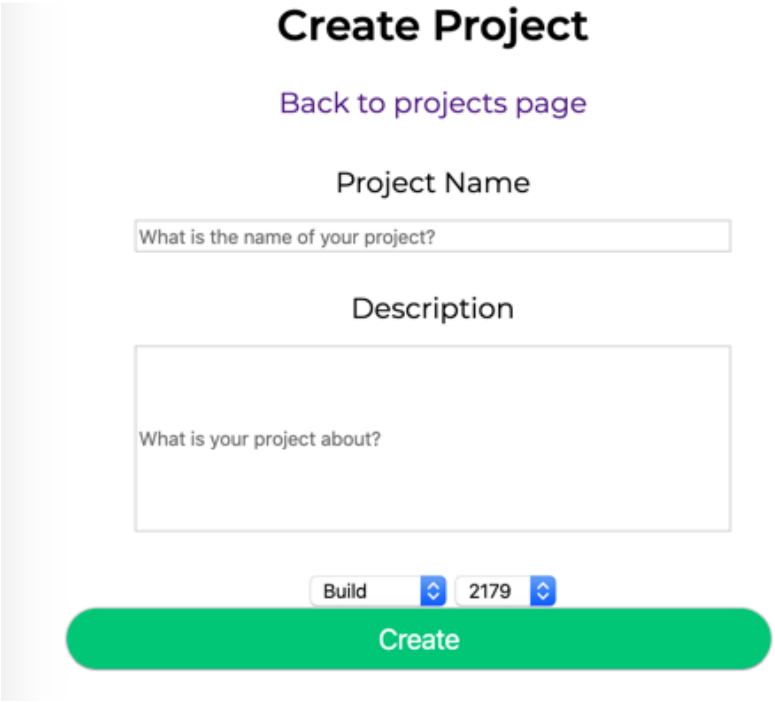

Figure 5.16 - shows the website frond-end view of the forms that were created using a combination of HTML, CSS & PHP.

5.8 - Conclusion

The implementation of this system is one of the most critical aspects of this project. It combines all of the previous steps from the requirements chapter and the design chapter. The implementation chapter is about turning the concepts of the project into a reality so that the system can be used in the real world. When implementing the system, the project developers realised that certain design concepts that were thought about in the design/requirement phases, could not actually be implemented. Some features looked different in the actual implementation, compared to the design. The aesthetic difference between the design and implementation mainly occurred because there was significantly more data loaded on the implemented page compared to the initial low and mid-fidelity designs. This aesthetic difference required the developers to change the design of the implementation to accommodate for the unforeseen design issues. In future projects, the project designers will consider how their designs will look once large amounts of data are added to the page. Considering this beforehand reduces the amount of changes that need to be made to the systems design when it comes to the implementation.

This chapter also discussed implementation issues that the project developers experienced. These issues were documented so that the same mistakes do not happen again in the future. All the technologies used by the team were also document so that that in future, if a similar project needs to be made, the team can refer to this chapter as a guide to what tools they will need to use. Implementing certain functions of the project caused difficulty and confusion amount the project team. Research by the developers was done to successful implement the different features. Code snippets of certain features were also included in this document. The code can be referred to again in future by the project developers as a point of reference if needing to build something similar in the future.

6 - Validation

6.1 - Introduction

The purpose of this chapter is to examine the various ways that elements of this system have been tested. In any project, it is essential to consider how the elements of the solution are going to be evaluated. Software components need to be tested in specific test cases to ensure that they everything works as intended. It is vital to test that the functional requirements for the system are working (CRUD functionality). Furthermore, essential to test if the non-functional requirements are working (loading times, responsiveness). This chapter will go into detail regarding how the following components of the system were tested: system requirements, system design and system implementation. The validation methods used will be praised and critiqued where necessary, in the hope of being able to use the testing experienced gaining in this project in future projects.

6.2 - Requirements

The systems requirements were validated between the client and the project developers. Several discussions were had between them so that the developers had a clear understanding of what the client wanted the system to do. The full list of functional requirements went through several iterations. Initially, the project developers had a brief meeting with the client to talk about the desired function of the system. The developers then went away and created a list of system requirements. However, the first few attempts of this list produced requirements that were not entirely in line with the client's vision of the solution.

The final list of functional requirements was created in a face-to-face meeting with the client and the development team. The two parties had a meeting in which they worked together to complete the list of functional and non-functional requirements. In future projects, the development team would preferably have this meeting with the client sooner so that the requirements of the project can be established earlier.

6.3 - Entity Relation Diagram

There were several design components of the system that needed to be validated. The first component that was tested was the ERD design for the database of the system. During the design of the ERD, SQL queries were made by the developers. These queries were made to ensure the ERD was designed in a way that allowed the system to fulfil the listed requirements. If it was not possible to create a query to execute a function on the requirements list, this was a sign that the design needed to be altered. The queries that were used to test the ERD design of the system can be found in **appendix I** of this report. The ERD went through multiple iterations to find the most efficient design to use in the system. Each design iteration was discussed with the client. This validation method proved to be useful because it forces the developers to look past the design

phase of the system. It forces developers to start thinking about the implementation process of creating SQL queries once the design has been translated into an actual database.

6.4 - Activity Diagram

The activity diagram was validated by speaking to the client of the project. In order to create the systems activity diagram, the developers referred to the requirements that were previously made. Firstly, the developers looked at the functional requirements of the project and thought of logical steps that the end-users of the systems could take to perform a task. After thinking about the flow of steps to perform tasks, the ideas were then translated into an activity diagram that shows all of the possible workflows that end-users of the stem could take. After translating the initial ideas into a diagram, it was then shown to the client of the project for feedback. The client recommended making a few changes to improve the workflow of the system. The developers then implemented these changes to meet the expectations of the client. Having the client validate the activity diagram proved to be useful to the development team. The validation helped the developers make sure that their vision of how the system should flow was in line with that of the clients.

6.5 - Sequence Diagram

The sequence diagram was validated by communicating with the client of the project. In order to create the systems sequence diagram, the developers referred to the requirements that were previously made. Firstly, the developers looked at the functional requirements of the project and thought of technology that would be needed so that end-users of the systems could perform the necessary system tasks. After thinking about the technology required, the ideas were then translated into a sequence diagram that shows all of the technological components that will be used to perform tasks in the system. After translating the initial ideas into a diagram, it was then shown to the client of the project for feedback. The client was happy with the logical of the sequence diagram. Having the client validate the sequence diagram proved to be useful to the development team. Tt helped the developers to make sure that their vision of what the technology architecture the system required was in line with the clients.

6.6 - Web Design

In order to create the website design of the system, wireframes were used to test design ideas. It is essential to use wireframes early on in the development process to establish the basic structure of a page before final design and contents are created. The wireframes provide a foundation from which the final solution can be built. When creating wireframes, the systems designers can identify user interface/user experience design flaws. Spotting design issues early in the design process is essential so that the final solution is as problem-free as possible. The wireframes were

shown to the client, who then gave the approval for those designs to be used for the full system implementation.

The system's web design was validated through usability tests that were carried out on five anonymous students at Kingston University (cannot be named due to privacy reasons). Performing a usability test on the system allows external people who do not have any involvement in the project to give feedback on the functionality and design of the system. It is useful to get the design of the system validated by external sources. This is because project externals would not have any prior knowledge of the project and can give a fresh, unbiased perspective to the team regarding the usability of the system. The usability test questionnaires can be found **Appendix J** of this document.

6.7 - Implementation

The systems implementation was validated using black-box testing. Black-box testing is a means of software testing that analyses the functionality of an application without inspecting its internal code structures (Black-box testing, 2020). This method of testing can be applied practically to every level of software testing. Some advantages of black-box testing are that tests are unbiased because the designer and tester work independently. The tester does not need any prior knowledge of specific programming languages to test the functionality of the system. Additionally, tests are conducted from a user's point-of-view and not of the developers. Test cases can also be devised quickly after the completion of specifications (What is Black Box Testing, 2020). For each function created for the system, a test was done to ensure that the functionality was working. If for whatever reason the function tested did not work, it was recorded in a black box testing table. Successful tests were also included in the tables too. To see a complete and thorough list of black-box table entries, please refer to **appendix H** of this document.

Test ID	Test Description	Expected Result	Actual Result	Status (Pass/Fail)
1	To see if the Projects webpage can successfully retrieve data from the backend database.	Data from the database should be returned to the webpage.	Data from the database was returned to the webpage.	Pass
2	To see if data from the Create Projects page can be sent to the database from the webpage.	After submitting data into a form on the webpage, the database should be updated with the new entry.	The form successfully added new data into the database.	Pass
3	Check if buttons in the nav bar direct to the correct pages when clicked on.	When clicking on specific buttons in the nav bar, user should be directed to the page referencing the title that they clicked on.	Some buttons redirected to the correct pages, however, not all buttons were linked to the correct pages.	Fail

Table 6.1 - Black-Box Testing Table (sample)

6.8 - Conclusion

Validating the various components of the system proved to be essential for meeting the client's executions of the system. Validating the system allowed the development team to get outside opinions and feedback on the work that they had been doing. It was through validating the project requirements that the developers discovered that their initial ideas regarding the functionality of the system, did not reflect what the client had in mind. Validating the design elements of the system with the client and potential users of the system proved to be essential to the project. This is because the development team were able to refine design elements and confirm that the design would be liked and usable to the targeted project stakeholders.

Black box testing also proved to be essential to the development of the system. Through black-box testing, the developers were able to ensure that all required functions of the system were properly working. Testing this ensured that users would not encounter any issues when using the website. All of the validation techniques used for various components proved to be useful. In future projects, developers can take the experience that they have gained from validating the components of this project and apply that experience (perhaps more efficiently) to future projects.

7 - Conclusion

7.1 - Introduction

This chapter will provide a concluding summary of this project and its outcomes. This chapter will review the original aims and objectives and whether or not they have been successfully met. This chapter will also provide a summary of pertinent strengths and limitations of the system and logical proposals for future work.

7.2 - Review of Project Aim

The overall aim of this project was to create a project deliverables management system (PDMS) with a web-based interface that is usable on both desktops and mobile devices. The intended implementation of the PDMS aimed to help both students and lecturers with the management of final-year projects. The system needed to provide a centralised location where all relevant information about a project and its relevant ongoing and completed tasks can be stored. This aim has been successfully met because front-end webpage and database have been successfully created, and integrated. The webpage that has been created allows data to be retrieved from to the database. The website also allows for data saved within the database to be manipulated by the user.

7.3 - Review of Project Objectives

Database

One objective for this project was to create a database that will be used to store relevant user information such as account details of both students and supervisors.

Outcome

- For the development of the database, an ERD model was created for the design of the database. The ERD was be made using an application named StarUML, which allows the creation of professional looking ERD database diagrams.
- After creating the ERD design, the database model was successfully created using phpMyAdmin.
- A data dictionary was created to collect descriptions of the data and objects in the database so that they could be viewed clearly in a single document.

Website

Another objective of this project was to develop a front-end, web-based application that was responsive to both desktop and mobile devices. The website successfully provides an interface in which the stakeholders of this project can use to manipulate and interact with data in the database.

Outcome

- Wireframes were successfully created to design the website.
- Based on the wireframes, a professional-looking, fully functional website was created for this project. The website allows users to interact with the backend database.
- The website has been successfully integrated into the backend database so that it is able to retrieve the relevant stored information and display it to the user on the webpage. The website can successfully store user input and send that back to the database.

7.4 - Critical Review

7.4.1 - Introduction

The critical review of this report will examine the overall development process of creating the Project Deliverables Management System (PDMS). The purpose of the PDMS was to create a web application that serves the purpose of allowing both students and supervisors to manage University projects. This chapter will discuss the management of this project and the design implementation method. This chapter will examine ways in which the development approach can be improved for future projects.

7.4.2 - Project Management Approach

The project management methodology that was chosen for the build of this project was the Agile methodology. The flexibility that the Agile method provided was exceedingly useful for this project as it was often necessary to alter deliverables that had been set from the development team. Often the due date of deliverables needed to be changed. This is because the completion of these deliverables sometimes took longer than expected.

The pushing back of due dates was applied mainly to implementation deliverables of the project. The implementation deliverables of the project were given the same amount of time to complete the project documentation and design elements. In hindsight, the implementation of deliverables should have been given *more* time to complete. More completion time was needed because these deliverables were more challenging to complete for the project team compared to others. Allowing extra time for the implementation deliverables to be completed would allow the project to run more efficiently. This is because deliverable dates would not frequently be changing, which makes it easier to plan future deliverables.

7.4.3 - Implementation

The styling of the front-end website of the system was manually styled using CSS. The developers of the system needed to take on additional learning so that they could successfully implement the required design elements from the wireframes. In hindsight, it may have been more time-effective if the development team opted to use an existing Bootstrap design. Bootstrap is a free and open-source CSS framework directed at responsive, mobile-first front-end web

development. It contains CSS- and JavaScript-based design templates for typography, forms, buttons, navigation, and other interface components. Using Bootstrap would have saved the development team time. This is because the developers were already familiar with Bootstrap and would not have had to do additional learning regarding styling CSS elements.

7.5 - Future Work

Email Reminders

For future development of this system, non-functional requirements can be examined. One of the non-functional requirements that has not been implemented into this system is the ability of the system to send email reminders to students and lecturers. The purpose of email reminders is to let users know that a project or deliverable is reaching its due date. Implementing this feature would reduce the probability of students and lecturers, forgetting that a deliverable needs to be completed.

Prioritising Tasks

Another non-functional requirement that has not been implemented is the ability for students and supervisors to prioritise deliverables that have been created. Being able to mark specific deliverables as 'important' would prove to be useful. It will be useful because, at a glance, students and supervisors would be able to see which tasks are most important to complete. This non-functional requirement could be implemented in future developments of this project.

Mobile Application

For future development of the system, a mobile application can be developed to run in conjunction with the website. A mobile application would provide increased ease of access to the system. This is because users will not need to type in the domain name of the website each time they would like to access the system. Instead, they could click on the application on their mobile device. Most modern websites have mobile applications, and for the expansion of this system, once the web application has been perfected, the next logical step is to create a mobile application.

7.6 - Personal Experience

The project development successfully overcame four main stages. Each stage created several challenges throughout the project. The first challenge came in the form of defining the project requirements with the client. This stage required several iterations of requirements to be created and prioritised. However, the project requirements were successfully agreed. As a result of this stage, I have gained experience in requirements elicitation and can use the experience if future projects that I am involved in.

The second stage was creating the database of the system. Designing the database proved to be challenging because there were several possible ways that the database could be designed. However, it was necessary to find the most *efficient* design so that queries could be run as quickly as possible. After creating the design of the database, it was then created in phpMyAdmin, and datasets were created to populate the tables. As a result of this stage, I have increased my SQL knowledge and would now consider myself to be competent in the relational database design and implementation.

The final stages were creating the front-end webpage of the system and also writing the project report. Developing the website required me to do some additional learning to improve my HTML, CSS, PHP and Javascript knowledge. However, as a result of this project, I believe my knowledge of all coding and scripting languages previously mentioned has drastically improved. I would now consider myself to be competent in all of them.

Overall this project has provided me with beneficial learning experiences. It has provided me with the chance to apply my database driven website knowledge to the field of software engineering. My prior experience with database-driven web development coding greatly helped me in this project. For this system, I have gone beyond my prior basic web development and database knowledge and have achieved results further than I could have ever imagined. My greatest achievement in this project was the creation of the front-end website and its ability to link to the phpMyAdmin database to create, read, edit and delete data (CRUD). I learnt a lot of valuable skills from this project which will have a huge impact on my future developments. This project has broadened my arisen in the software development field.

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Appendix A – Low Fidelity Wireframes

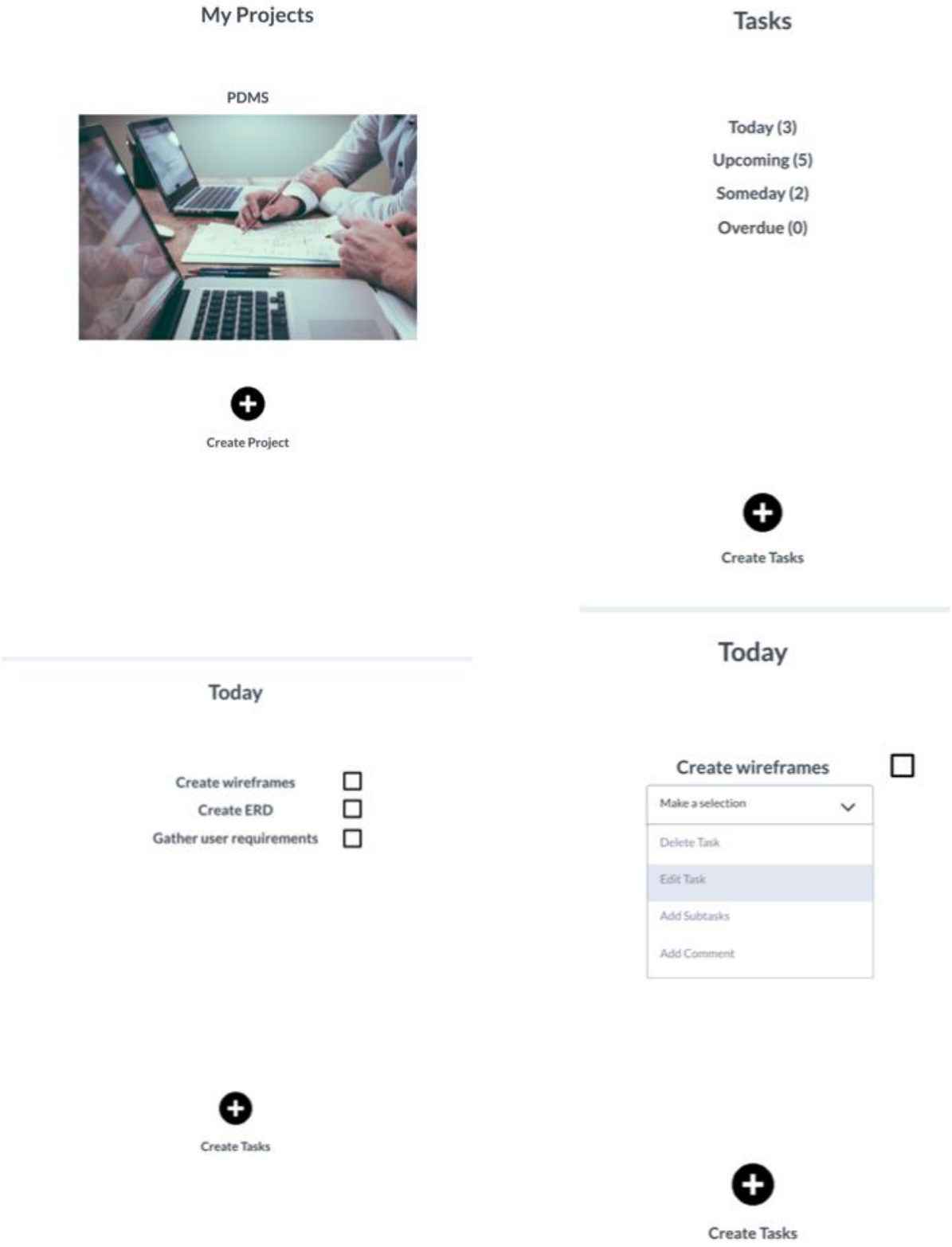


Figure A.1 Students Website View

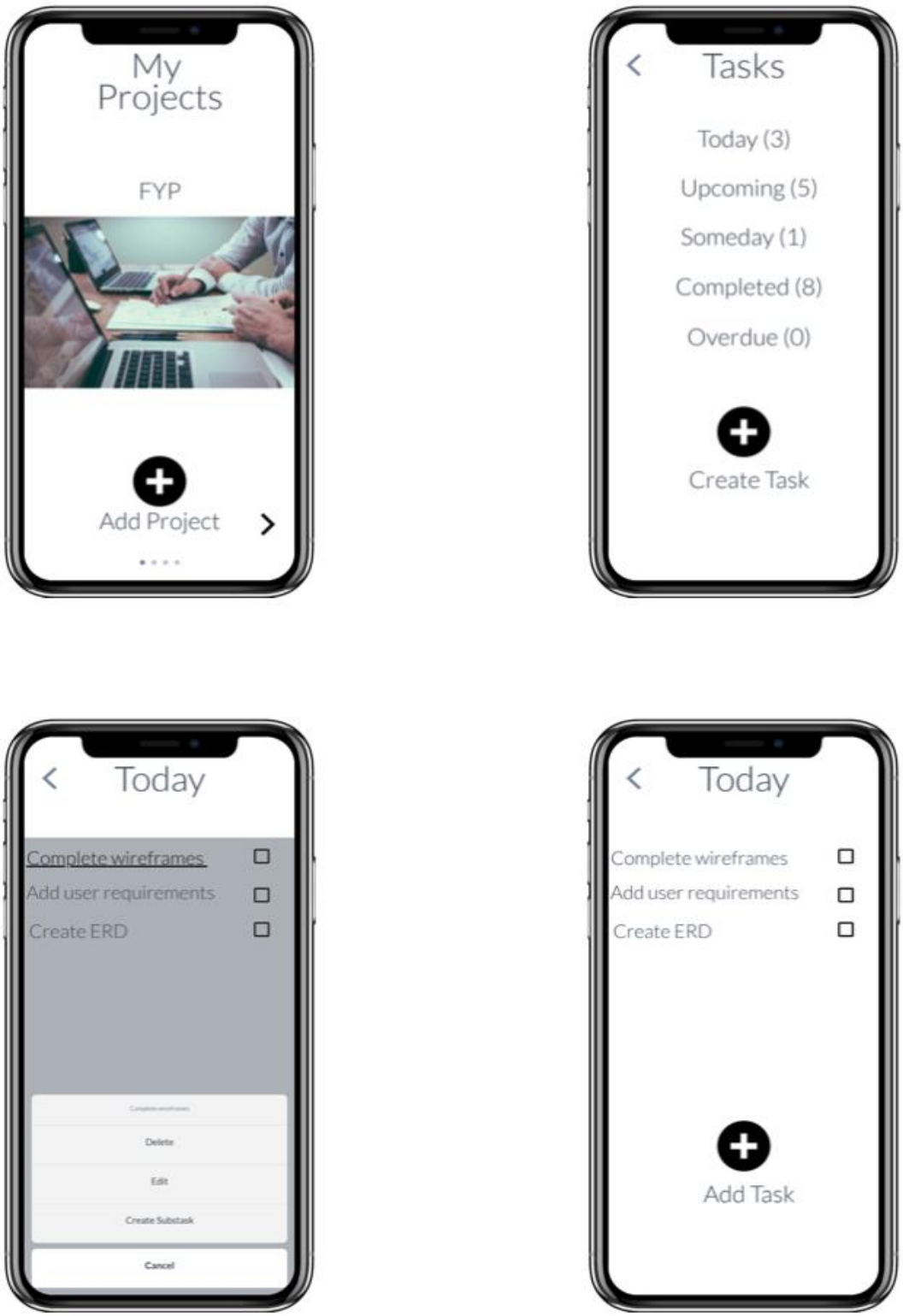


Figure A.2 Mobile Website View

Appendix B – Project Deliverable Plan

Deliverable List			
Day	*Date*	*Deliverable*	*All*
Fri	9/27/19	Documentation	Proposal Appendices
Fri	10/4/19	Methodology	Initial ERD (Initial data structures)
Fri	10/4/19	Research	Portfolio of Past Dissertations
Fri	10/4/19	Documentation	Complete draft proposal
Fri	10/4/19	Methodology	SWOT / Risk Analysis
Fri	**10/11/19**	**Documentation**	**Project Proposal**
Fri	10/11/19	Methodology	Client signed off use cases
Fri	10/11/19	Implementation	Install development platform
Fri	10/18/19	Methodology	Stakeholder analysis
Fri	10/18/19	Documentation	TOC Requirements Chapter
Fri	10/18/19	Implementation	Establish PHP connection to database
Fri	10/25/19	Research	Volume of testing data required
Fri	10/25/19	Methodology	Use Cases and User Stories
Fri	10/25/19	Implementation	Simple SQL and PHP report
Fri	11/1/19	Methodology	List of Requirements / MoSCoW
Fri	11/1/19	Research	Identify Review Topic
Fri	11/1/19	Implementation	Simple Form with SQL insert
Fri	11/8/19	Documentation	Draft Requirements Chapter
Fri	11/8/19	Research	Bootstrap/Mobile API Interface
Fri	11/8/19	Implementation	Inner Join report
Fri	11/15/19	Methodology	Sketches
Fri	11/15/19	Documentation	TOC Design Chapter
Fri	11/15/19	Research	Identify SQL queries from requirements
Fri	11/22/19	Implementation	Implement mockups of key webpages
Fri	11/22/19	Methodology	Sequence diagrams
Fri	11/22/19	Research	Plan for prototype demonstration
Fri	11/29/19	Documentation	Updated Requirements Chapter
Fri	11/29/19	Implementation	Key report 1
Fri	11/29/19	Methodology	Revised ERD and Data Dictionary
Fri	12/6/19	Implementation	Key form 1
Fri	12/6/19	Research	Test Strategy
Fri	12/13/19	Documentation	Draft Design Chapter
Fri	**12/13/19**	**Presentation**	**Early Prototype demonstration**
Fri	1/10/20	Methodology	Wireframes/Navigation
Fri	1/10/20	Research	Key References and Mind Map
Fri	1/10/20	Implementation	Key report 2
Fri	1/17/20	Documentation	Updated Design Chapter
Fri	1/17/20	Research	Data generation plan
Fri	1/17/20	Implementation	Key form 2
Fri	1/24/20	Research	Identify most complex SQL
Fri	1/24/20	Implementation	Key form 3
Fri	1/24/20	Implementation	Key report 3
Fri	1/24/20	Research	Identify viva story
Fri	1/31/20	Documentation	Draft Review Chapter
Fri	1/31/20	Implementation	Key report 4
Fri	1/31/20	Implementation	Key form 4
Fri	1/31/20	Evidence	Populate database with required data
Fri	2/7/20	Documentation	TOC Implementation Chapter
Fri	2/7/20	Documentation	Technologies and Architecture Section
Fri	2/7/20	Implementation	Key report 5
Fri	2/7/20	Implementation	Key form 5
Fri	2/14/20	Evidence	Test Tables
Fri	2/14/20	Documentation	Second Major Section
Fri	2/14/20	Implementation	Key report 6
Fri	2/14/20	Implementation	Key form 6
Fri	2/21/20	Research	Identify viva-oriented functionality

Fri	11/1/19	Research	Identify Review Topic
Fri	11/1/19	Implementation	Simple Form with SQL insert
Fri	11/8/19	Documentation	Draft Requirements Chapter
Fri	11/8/19	Research	Bootstrap/Mobile API Interface
Fri	11/8/19	Implementation	Inner Join report
Fri	11/15/19	Methodology	Sketches
Fri	11/15/19	Documentation	TOC Design Chapter
Fri	11/15/19	Research	Identify SQL queries from requirements
Fri	11/22/19	Implementation	Implement mockups of key webpages
Fri	11/22/19	Methodology	Sequence diagrams
Fri	11/22/19	Research	Plan for prototype demonstration
Fri	11/29/19	Documentation	Updated Requirements Chapter
Fri	11/29/19	Implementation	Key report 1
Fri	11/29/19	Methodology	Revised ERD and Data Dictionary
Fri	12/6/19	Implementation	Key form 1
Fri	12/6/19	Research	Test Strategy
Fri	12/13/19	Documentation	Draft Design Chapter
Fri	**12/13/19**	**Presentation**	**Early Prototype demonstration**
Fri	1/10/20	Methodology	Wireframes/Navigation
Fri	1/10/20	Research	Key References and Mind Map
Fri	1/10/20	Implementation	Key report 2
Fri	1/17/20	Documentation	Updated Design Chapter
Fri	1/17/20	Research	Data generation plan
Fri	1/17/20	Implementation	Key form 2
Fri	1/24/20	Research	Identify most complex SQL
Fri	1/24/20	Implementation	Key form 3
Fri	1/24/20	Implementation	Key report 3
Fri	1/24/20	Research	Identify viva story
Fri	1/31/20	Documentation	Draft Review Chapter
Fri	1/31/20	Implementation	Key report 4
Fri	1/31/20	Implementation	Key form 4
Fri	1/31/20	Evidence	Populate database with required data
Fri	2/7/20	Documentation	TOC Implementation Chapter
Fri	2/7/20	Documentation	Technologies and Architecture Section
Fri	2/7/20	Implementation	Key report 5
Fri	2/7/20	Implementation	Key form 5
Fri	2/14/20	Evidence	Test Tables
Fri	2/14/20	Documentation	Second Major Section
Fri	2/14/20	Implementation	Key report 6
Fri	2/14/20	Implementation	Key form 6
Fri	2/21/20	Research	Identify viva-oriented functionality
Fri	2/21/20	Documentation	Updated Review Chapter
Fri	2/21/20	Evidence	Simple reports utilising big data
Fri	2/28/20	Implementation	Login functionality
Fri	2/28/20	Documentation	Draft Implementation Chapter
Fri	3/6/20	Implementation	Dashboard 1
Fri	3/6/20	TBC	TBC
Fri	3/13/20	Evidence	Inner joins utilising big data
Fri	3/13/20	Documentation	Draft Validation Chapter
Fri	3/20/20	Implementation	Dashboard 2
Fri	3/20/20	Documentation	Draft Introduction and Conclusion
Fri	3/20/20	Research	Revised viva-oriented functionality
Fri	3/27/20	Documentation	Appendices and references
Fri	4/3/20	Presentation	Practice Vivas
Fri	4/3/20	Implementation	Dashboard 3
Fri	4/10/20	Documentation	Complete draft report (and submitted)
Fri	4/17/20	Implementation	Last viva functionality implemented
Fri	4/24/20	Presentation	Practice Vivas
Mon	**4/27/20**	**Documentation**	**Report Submission**

Figure B.1 Project Deliverables Plan

Appendix C - Data Discussion

C.1 Introduction

The volume of data that the PDMS is anticipated to manage has been discussed with the Client. The database for this system is likely to expand regarding the entries in the system over time significantly. One must consider the potential technical malfunctions that can happen alongside the need to keep all of the data secure. This chapter will examine a plethora of potential issues that need to be considered, such as data security, scaling complexities and also the potential amount of total data that will be stored within the specific database tables. Adding significantly more users to the database will have an impact on the speed in which queries can be run and how quickly the results of the queries are returned.

C.2 Scalability

Users

It can be assumed that the number of students using this system is likely to increase over five years; therefore, the scalability of the PDMS needs to be considered for the future. In the first year of usage of the system, one can assume that there would be 4000 students in total stored within the database. An assumption can be made that there will be on average, four new users added to the database every week for the next five years. This increase in users results in an additional 1040 students added to the system and therefore, resulting in a large increase of total student users on the system.

Supervisors

It can be assumed that a single university could have 100 project supervisors using the PDMS for the first year. Each of these supervisors would have their information added to the database. The number of supervisors on the system is likely to increase over five years if the software grows in popularity and reputation. This could mean that other universities start to adopt the system. If the system was being used by 50 different universities within five years, the total amount of supervisors on the system would be 5000.

Projects

Each student in the system will be able to create an unlimited amount of projects. It can be assumed that each student may have a total of 5 - 10 projects. If each student creates 10 projects, this will mean around 20,000 projects will be stored in the database.

Comments

It will be possible for supervisors to comment on a project. Each project can have an unlimited amount of comments. If each project averaged around 5 comments, the total amount of comments stored in the database would be approximately 100,000.

Users Creating Tasks

In this system, students will be able to create an unlimited amount of project tasks which then will be stored in the database. If each student has at least 100 tasks in total, this will result in over 200,000 data entries being stored. Supervisors will also be able to create tasks for students. If each supervisor has an average of 5 students and sets them each at least 5 tasks there are 100 supervisors, there will be an additional 2500 tasks.

Checklist

It will be possible for each task in a project to have checklist within. Each task could have multiple items on the checklist. If each of the 100 tasks in the table had three checklist items, there would be 300 checklist items stored in the database per student. With 4000 students this would mean that there would be 1,200,000 checklist items in the database.

Completed Tasks

The PDMS will store completed tasks in a logbook format so that users can look back at things they have previously done. If a user completes 100 tasks every year for five years, the total number of completed tasks will be 500. If you multiple this by 4000 total students using the system, there would be 2,00,000 completed tasks stored in the database.

Deleted Tasks

The PDMS will store deleted tasks in a trash category so that users can look back at things they have previously deleted. If a user deletes 10 tasks every year for five years, the total number of completed tasks will be 50. If you multiple this by the number of (potential) total students using the system after five years, there would be 100,000 deleted tasks.

Evidence

For specific project deliverables, students may be required to upload images as proof of completion. The uploaded image then needs to be stored in a database. Images take up significantly more storage than text and therefore, the number of images, and the size of images users can upload has been examined. The purpose of the image upload is to allow the supervisor to check that work has been completed. If each of the 4000 users uploaded 3 images each per project, assuming that they had 5 projects, there would be 60,000 projects stored in the database.

ERD Tables	Rows Needed After Five Years
Users	5000
Supervisors	5000
Projects	20,000
Comments	100,000
Tasks	200,000
Checklist	1,200,000
Completed Tasks	2,000,000
Deleted Tasks	100,000
Evidence	60,000

Table C1 – Data Entries After Five Years

Solution

This reduction in speed can cause the system to feel slow when being used, and the issue will only magnify as more people start to use the system. A way that this issue can be mitigated is to invest in a server with the required capacity to store lots of data. A server with an appropriate amount of storage space will allow for the system to be significantly scaled over time without any noticeable reductions in performance. Another way to ensure that there is enough space available on the server is by frequently deleting old data entries that are no longer required. After five years, user details on the system will be removed in order to make space for new users. The reason for allowing a five-year time period is because, after this time, students (likely to be former students) will not need to access this data. Therefore, there is no need for it to take up space in the database.

Security

The system will be storing information about university students and supervisors; therefore, it is essential to follow GDPR. GDPR takes privacy law to account for more recent technical developments and how people use data. It increases restrictions on what organisations can do with user data, and it extends the rights of individuals to access and control data about them. If a user requests for their data to be removed from the system, by law, it would need to be removed. A data breach typically costs a company million in fines, as well as a loss of reputation and goodwill. This project will mitigate the risk of breaching GDPR because the system will be using fictitious open-source datasets which are available on the internet and accessible to anyone. Some data, such as the project deliverables for specific students, will need to be manually entered during the development of the system. This project will not be putting anyone's real personal information at risk because the datasets used will be fictitious and open source.

Appendix D - Use Case Description Catalogues

ID	Name	Actor(s)	Complexity	Priority
UC1	Upload Evidence	Student	High	7
UC2	Submit Work	Student	High	7
UC3	Create Deliverables	Student & Supervisor	High	10
UC4	Edit Deliverables	Student & Supervisor	High	10
UC5	Delete Deliverables	Student & Supervisor	High	10
UC6	Check Deliverables	Student & Supervisor	High	10
UC7	Login/Logout	Student & Supervisor	High	8
UC8	View Grades	Student & Supervisor	Med	5
UC9	Submit Feedback	Supervisor	Med	6
UC10	View Statistics	Supervisor	Low	5
UC11	View Student	Supervisor	High	8
UC12	View Evidence	Supervisor	High	7

Catalogue 1

Use Case ID	UC1
Use Case Name	Upload Evidence
Description	Allows the student to upload evidence of completed work
Actor(s)	Student
Frequency of Use	Will vary, anywhere from daily to never
Triggers	Supervisor wants to see proof of work
Preconditions	Student has access to internet and domain
Flow of Events	Log in to system, click on project, select evidence, then upload evidence
Exception Flow	The site is down for maintenance

Catalogue 2

Use Case ID	UC2
Use Case Name	Submit Work
Description	Allows the student to upload documents of completed work
Actor(s)	Student
Frequency of Use	Will vary, anywhere from daily to never
Triggers	Supervisor wants to see proof of work
Preconditions	Student has access to internet and domain
Flow of Events	Log in to system, click on project, select submit, then upload document
Exception Flow	The site is down for maintenance

Catalogue 3

Use Case ID	UC3
Use Case Name	View Supervisor
Description	View allocated supervisor
Actor(s)	Student
Frequency of Use	Rare, probably only a couple of times
Triggers	Wants to know who supervisor is
Preconditions	Student has access to internet and domain
Flow of Events	Log in to system, click on project, select project details, supervisor is shown
Exception Flow	The site is down for maintenance

Catalogue 4

Use Case ID	UC4
Use Case Name	Create Deliverables
Description	Lets users create tasks
Actor(s)	Student & Supervisor
Frequency of Use	Weekly
Triggers	Want to set a reminder for a task to complete
Preconditions	User has access to internet and domain

Flow of Events	Log in to system, click on project, create task
Exception Flow	The site is down for maintenance

Catalogue 5

Use Case ID	UC5
Use Case Name	Edit Deliverables
Description	Lets users edit tasks
Actor(s)	Student & Supervisor
Frequency of Use	Weekly
Triggers	Wants to change a task
Preconditions	User has access to internet and domain
Flow of Events	Log in to system, click on project, edit task
Exception Flow	The site is down for maintenance

Catalogue 6

Use Case ID	UC6
Use Case Name	Delete Deliverables
Description	Lets users delete tasks
Actor(s)	Student & Supervisor
Frequency of Use	Monthly
Triggers	Wants to remove a task
Preconditions	User has access to internet and domain
Flow of Events	Log in to system, click on project, delete task
Exception Flow	The site is down for maintenance

Catalogue 7

Use Case ID	UC7
Use Case Name	Check Deliverables
Description	Lets users see tasks
Actor(s)	Student & Supervisor
Frequency of Use	Weekly
Triggers	Wants to see what tasks need to be done
Preconditions	User has access to internet and domain

Flow of Events	Log in to system, click on project, tasks appear
Exception Flow	The site is down for maintenance

Catalogue 8

Use Case ID	UC8
Use Case Name	Login/Logout
Description	Sign in and out of accounts
Actor(s)	Student & Supervisor
Frequency of Use	Daily
Triggers	Wants to access the system
Preconditions	User has access to internet and domain
Flow of Events	Visit domain, enter email and password
Exception Flow	The site is down for maintenance

Catalogue 9

Use Case ID	UC9
Use Case Name	View Grades
Description	Lets users see grades for a project
Actor(s)	Student & Supervisor
Frequency of Use	Monthly
Triggers	Want to see what grade has been given for a project
Preconditions	User has access to internet and domain
Flow of Events	Log in to system, click on project, click on grades
Exception Flow	The site is down for maintenance

Catalogue 10

Use Case ID	UC10
Use Case Name	Submit Feedback
Description	Lets supervisors comment on projects
Actor(s)	Supervisor
Frequency of Use	Weekly

Triggers	Wants to inform student about project
Preconditions	Supervisor has access to internet and domain
Flow of Events	Log in to system, click on project, submit feedback
Exception Flow	The site is down for maintenance

Catalogue 11

Use Case ID	UC11
Use Case Name	View Statistics
Description	Lets supervisor see who is on or off track when completing deliverables
Actor(s)	Supervisor
Frequency of Use	Monthly
Triggers	Wants to see who is on or off track
Preconditions	Supervisor has access to internet and domain
Flow of Events	Log in to system, click view students, see statistics dashboard
Exception Flow	The site is down for maintenance

Catalogue 12

Use Case ID	UC12
Use Case Name	View Student
Description	Lets supervisor see all their students
Actor(s)	Supervisor
Frequency of Use	Monthly
Triggers	Wants to see what students they supervise
Preconditions	Supervisor has access to internet and domain
Flow of Events	Log in to system, click view students
Exception Flow	The site is down for maintenance

Catalogue 13

Use Case ID	UC13
Use Case Name	View Evidence
Description	Lets supervisor see upload work evidence
Actor(s)	Supervisor
Frequency of Use	Will vary, anywhere from daily to never
Triggers	Wants to see if work has definitely been completed
Preconditions	Supervisor has access to internet and domain
Flow of Events	Log in to system, click projects, click on view evidence
Exception Flow	The site is down for maintenance

Appendix E - Data Dictionary

Table A.1 - Projects

Field Name	Data Type	Character Length	Description	Example
ProjectID PK	Intager	5	Unique table identifier. Primary Key.	1
ProjectName	Varchar2	255	The title of the project	Management Website
ProjectDescription	Varchar2	255	The description of the project	To build a management description
StudentID FK	Intager	5	The unique identifier of the students table	10
GradeID FK	Varchar2	1	The unique identifier of the grades table	A
ProjectType FK	Varchar2	30	The unique identifier of the project type table	Build
SupervisorID FK	Intager	5	The unique identifier of the supervisor table	22

Table A.2 Supervisors

Field Name	Data Type	Character Length	Description	Example
SupervisorID PK	Intager	5	Unique table identifier. Primary Key.	1
FirstName	Varchar2	255	Supervisor's first name	Bob
Surname	Varchar2	255	Supervisor's surname	Smith

Email	Intager	5	The supervisors email address	one@one.com
PhoneNumber	Varchar2	1	The supervisors mobile number	4473848263

Table A.3 - Students

Field Name	Data Type	Character Length	Description	Example
StudentID PK	Intager	5	Unique table identifier. Primary Key.	1
FirstName	Varchar2	255	Students's first name	Bob
Surname	Varchar2	255	Students surname	Smith
Email	Intager	5	The students email address	one@one.com
PhoneNumber	Varchar2	1	The students mobile number	4473848263
SupervisorID FK	Intager	5	Foreign key from the supervisors table	1

Table A.4 - Comments

Field Name	Data Type	Character Length	Description	Example
CommentID PK	Intager	5	Unique table identifier. Primary Key.	1
Comment	Varchar2	255	The text that makes up a comment	The work is really good!
ProjectID FK	Varchar2	255	Foreign key from the projects table	22

Table A.5 - Evidence

Field Name	Data Type	Character Length	Description	Example
EvidenceID PK	Intager	5	Unique table identifier. Primary Key.	1
Evidence	BLOB		The text that makes up a comment	Evidence.png
ProjectID FK	Varchar2	255	Foreign key from the projects table	22

Table A.6 - Tasks

Field Name	Data Type	Character Length	Description	Example
TaskID PK	Intager	5	Unique table identifier. Primary Key.	1
Description	Varchar2	255	Information about a task	Create wireframes
DueDate	Date	n/a	Shows when a task is due for completion.	01/01/2020
ProjectID FK	Varchar2	255	Foreign key from the projects table	22

Table A.7 - Subtask

Field Name	Data Type	Character Length	Description	Example
SubTaskID PK	Intager	5	Unique table identifier. Primary Key.	1
Description	Varchar2	255	Information about a task	Create wireframes
TaskID FK	Intager	5	Foreign key from the tasks table	5

Table A.8 - TaskStatus

Field Name	Data Type	Character Length	Description	Example
Status PK	Varchar2	50	Unique table identifier. Primary Key.	Completed

Table A.9 - ProjectType

Field Name	Data Type	Character Length	Description	Example
TypeID PK	Varchar2	50	Unique table identifier. Primary Key.	1
Description	Varchar2	255	Description of the project type	Create wireframes

Table A.10 - Grade

Field Name	Data Type	Character Length	Description	Example
GradeID PK	Varchar2	1	Unique table identifier. Primary Key.	A
Description	Varchar2	255	Description of what the grade means	The work has been completed to an excellent standard.

Appendix F – Sketches

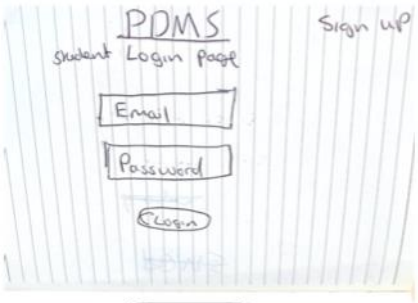


Figure B.1

Figure B.2

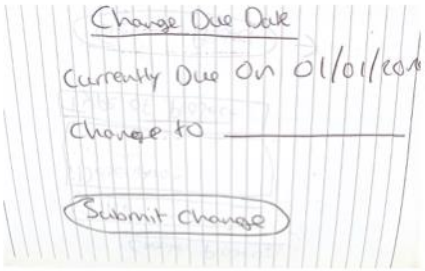


Figure B.3

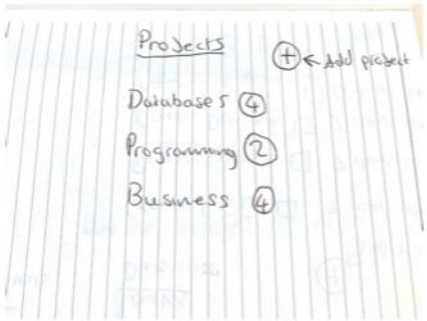


Figure B.4

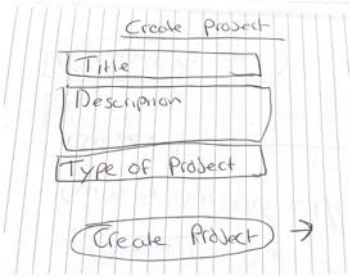


Figure B.5

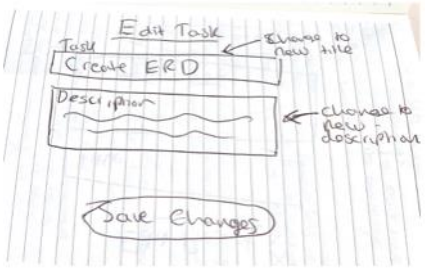


Figure B.6

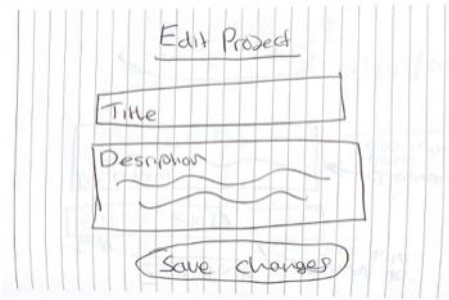


Figure B.7

Appendix G - Mid-Fidelity Wireframes

Figure C.1 - Login

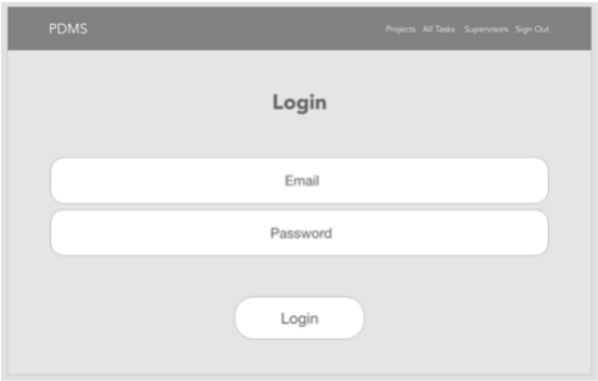


Figure C.2 - Projects

Figure C.3 - Project Details

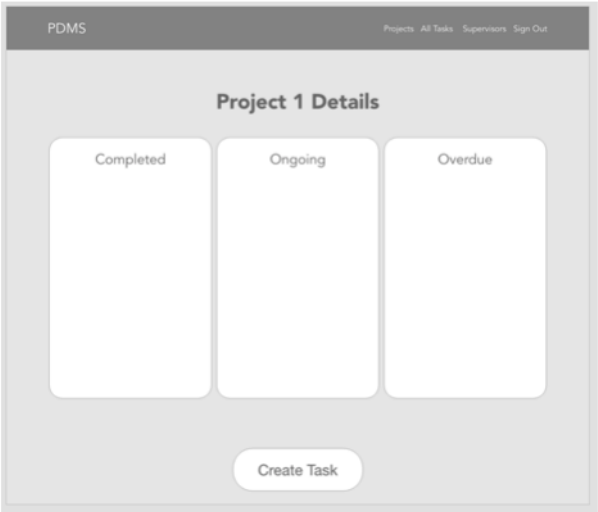


Figure C.4 - Supervisor Details

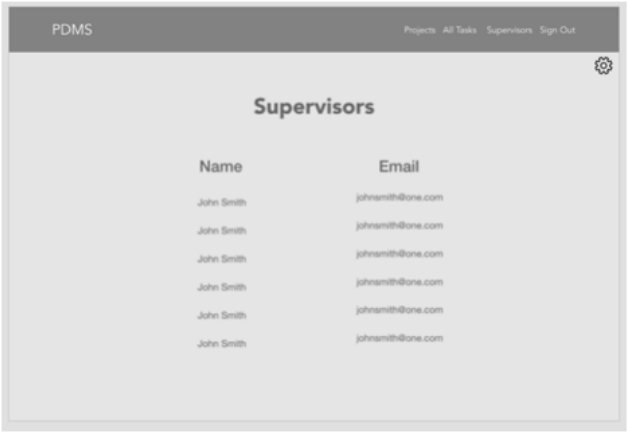


Figure C.5 - Create Project

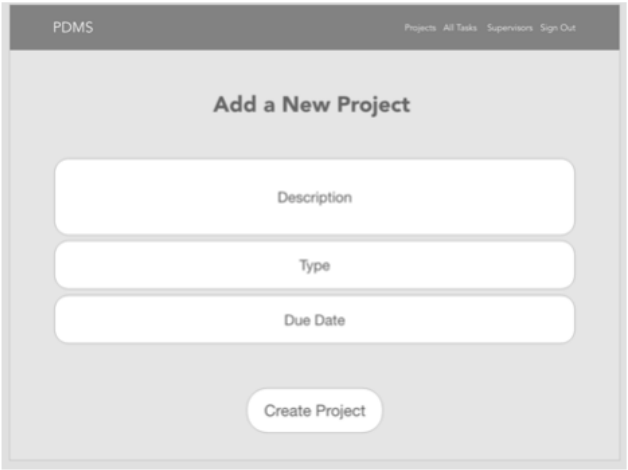
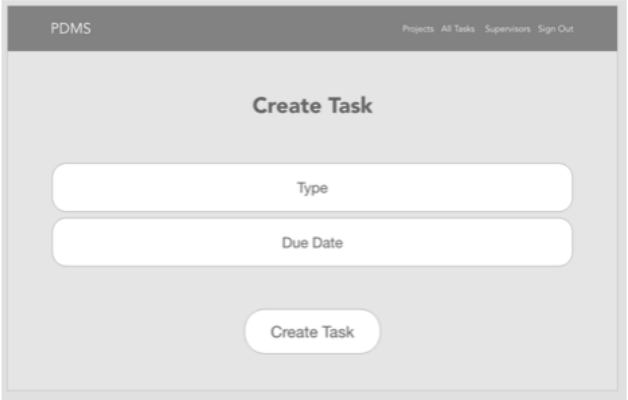


Figure C.6 - Create Task



Appendix H – Black-Box Testing Tables

Test ID	Test Description	Expected Result	Actual Result	Status (Pass/Fail)
1	To see if the Projects webpage can successfully retrieve data from the backend database.	Data from the database should be returned to the webpage.	Data from the database was returned to the webpage.	Pass
2	To see if data from the Create Projects page can be sent to the database from the webpage.	After submitting data into a form on the webpage, the database should be updated with the new entry.	The form successfully added new data into the database.	Pass
3	Check if buttons in the nav bar direct to the correct pages when clicked on.	When clicking on specific buttons in the nav bar, user should be directed to the page referencing the title that they clicked on.	Some buttons redirected to the correct pages, however, not all buttons were linked to the correct pages.	Fail
4	To see if the Tasks webpage can successfully retrieve data from the backend database.	Data from the database should be returned to the webpage.	Data from the database was returned to the webpage.	Pass
5	To see if the SubTasks webpage can successfully retrieve data from the backend database.	Data from the database should be returned to the webpage.	Data from the database was returned to the webpage.	Pass
6	To see if data from the Create Tasks page can be sent to the database from the webpage.	After submitting data into a form on the webpage, the database should be updated with the new entry.	The form successfully added new data into the database.	Pass
7	To test if the Create Task button successfully navigates to the the correct page.	The button should navigate to the correct page.	The button navigated to the correct page.	Pass

8	To test if the edit task button allows a task to be edited.	The button should prompt the user to change edit the task.	System error.	Fail
9	Date javascript function. When clicking on a form, a calendar showing the month and date should show.	The form should display a date calendar.	The system successfully displayed a date calendar.	Pass
10	Login password form should be stared out.	When typing into the password form, the password should not be visible.	Text in the password form is stared out and is not visible.	Pass
11	Website should be responsive and should resize when stretched from a mobile view to a desktop view.	Components of the website should be styled neatly and should be more spaced out.	Components of the website adjusted appropriately.	Pass
12	Website should be responsive and should resize when stretched from a mobile view to a tablet view.	Components of the website should be styled neatly and should be more spaced out.	Components of the website did not adjust appropriately.	Fail
13	Test if supervisor page is functioning as expected.	Supervisor page should return a list of names and email addresses address of supervisor users from the database.	Page successfully returned supervisor information in a card format.	Pass
14	View uploaded evidence.	User should be able to see evidence that has previously been uploaded.	Evidence files stored in the database are shown.	Pass
15	Upload evidence.	User should be able to upload files as evidence of work completed.	Evidence file did not successfully upload to the database.	Fail
16	Login functionality.	Using a specific user name and password should allow a user to login and see a specific view of the system	Login failed, error in the code.	Fail

		relating to the login details entered.		
17	Delete Project.	User should be able to delete a project from the database.	The project did not delete from the database.	Fail
18	Delete Task.	User should be able to delete a task from the database.	The task did not delete from the database.	Fail
19	Delete Subtask.	User should be able to delete a subtask from the database.	The subtask did not delete from the database.	Fail
20	Delete Evidence	User should be able to delete a piece of evidence from the database.	The evidence did not delete from the database.	Fail
21	Hamburger menu in mobile view should display nav bar menus.	Upon clicking on the hamburger menu, a navigation bar should appear on the page when in a mobile view.	Nothing happens when clicking on the hamburger menu.	Fail
22	PDMS logo at the top left of the page should direct the user to the homepage of the system.	Upon clicking got the button, user should be brought to the homepage.	The button is not linked to anything, therefore, clicking on the button does not do anything.	Fail
23	Sign out function.	Upon clicking on the sign out button, the user should be signed out and brought back to the landing page, from there they should be able to sign in again.	Click on the button signs the users out.	Pass
24	Inserting user data into the database.	Upon uploading a CSV file containing hundreds of user details, the database should save this information and the frontend webpage	Dataset was successfully uploaded and webpage is able to retrieve the data.	Pass

		should able to display the details.		
25	Inserting project data into the database.	Upon uploading a CSV file containing project details, the database should save this information and the frontend webpage should able to display the details.	Dataset was successfully uploaded and webpage is able to retrieve the data.	Pass
26	Inserting tasks data into the database.	Upon uploading a CSV file containing task details, the database should save this information and the frontend webpage should able to display the details.	Dataset was successfully uploaded and webpage is able to retrieve the data.	Pass
27	Inserting subtasks data into the database.	Upon uploading a CSV file containing subtask details, the database should save this information and the frontend webpage should able to display the details.	Dataset was successfully uploaded and webpage is able to retrieve the data.	Pass
28	Adding a comment.	User should be able to add a comment to a task.	Comment was successfully added to a task.	Pass
29	View a comment.	Upon clicking on a task, the user should be able to see all comments that have been created for the task.	Comments are displayed when a user clicks on a task.	Pass

31	View supervisor information.	When clicking on a supervisor card on the supervisor page, full details about the supervisor should be displayed.	Syntax error occurred and expected results were not shown.	Fail
32	To test if the edit task button allows a task to be edited.	The button should prompt the user to change edit the task.	System error.	Fail
33	To test if the delete project button allows a project to be edited.	The button should delete the selected project.	Project removed upon clicking delete.	Pass
34	To test if the complete button functionality is working correctly.	After clicking the 'completed' button on the tasks page, the status of the task should change to completed.	The button did not change the status of the task.	Fail
35	To test if the complete button functionality is working correctly.	After clicking the 'completed' button on the tasks page, the status of the task should change to completed.	The button updated the status of the task to 'completed'. Issue was that $_POST method was being used instead of $_GET method.	Pass

Appendix I – Test Queries

Requirements	SQL Queries
• As a student I would like to be able to view ongoing projects.	SELECT ProjectName FROM Projects WHERE ProjectStatus = 'Ongoing' AND StudentID = myStudentID;
• As a student I want to be able to see what tasks need to be completed for a project.	SELECT Description, Projects.ProjectName FROM Tasks INNER JOIN Projects ON Tasks.ProjectID = Projects.ProjectID WHERE Projects.ProjectID = 'myProjectID'
• As a student I would like to be able to view different tasks based on outstanding, late or completed.	SELECT tastDescription FROM Task WHERE TaskStatus = 'Ongoing' AND StudentID = myStudentID; SELECT tastDescription FROM Task WHERE TaskStatus = 'Late' AND StudentID = myStudentID; SELECT tastDescription FROM Task WHERE TaskStatus = 'Completed' AND StudentID = myStudentID;
• As a student I want to be able view my supervisor and see their contact details.	SELECT SupervisorName, Email FROM Supervisors WHERE SupervisorID = mySupervisorID;
• As a lecturer I would like to be able to view individual student projects.	SELECT ProjectName, FirstName FROM Projects INNER JOIN Students ON Projects.StudentID = Students.StudentID WHERE Students.StudentID = 'myStudentID'
• As a lecturer I would like to be able to identify which students had outstanding tasks	SELECT TaskDescription, DueDate, Students.FirstName, ProjectName FROM Tasks INNER JOIN Projects ON Tasks.ProjectID = Projects.ProjectID INNER JOIN Students ON Students.StudentID = Projects.StudentID
• As a student I want to be able view the grade given for a project.	SELECT ProjectName, ProjectDescription, Grade FROM Project WHERE StudentID = myStudentID;
• As a student I would like to be able to view the due date of a task.	SELECT taskDescription, DueDate FROM Tasks WHERE StudentID = myStudentID;

• As a lecturer I would like to be able to view the due date of a project.	SELECT ProjectDescription, DueDate FROM Projects;
• As a lecturer I want to be view the priority of a task.	SELECT taskDescription, Priority FROM Projects;
• As a student I would like to be able to view a subtask.	SELECT subtaskName FROM subtasks WHERE StudentID = myStudentID
• As a student I would like to be able to view the due date of a project.	SELECT ProjectDescription, DueDate FROM Projects WHERE StudentID = myStudentID;
• As a student I want to be able view full project description and details.	SELECT ProjectName, ProjectDescription, DueDate FROM Projects WHERE StudentID = myStudentID;
• As a lecturer I would like to see specific information about Students.	SELECT StudentName, Email, PhoneNo. FROM Students;

Appendix J – Completed Usability Test Questionnaires

This questionnaire was completed by an anonymously student at Kingston University. For privacy reasons, their identity will not be revealed.

Participant 1

How would you describe your overall experience with the website?	Good, the system was easy to use I seems useful.
What did you like the most about using this website?	I liked the design of the website. Nice colours and modern design elements.
What did you like the least?	Maybe the font could be changed.
What, if anything, surprised you about the experience?	Nothing surprised me.
What, if anything, caused you frustration?	Nothing caused frustration.

Participant 2

How would you describe your overall experience with the website?	It was easy to use.
What did you like the most about using this website?	I liked the simplicity of the website.
What did you like the least?	The colours.
What, if anything, surprised you about the experience?	Nothing.
What, if anything, caused you frustration?	The colours used.